The University of Melbourne
School of Mathematics and Statistics
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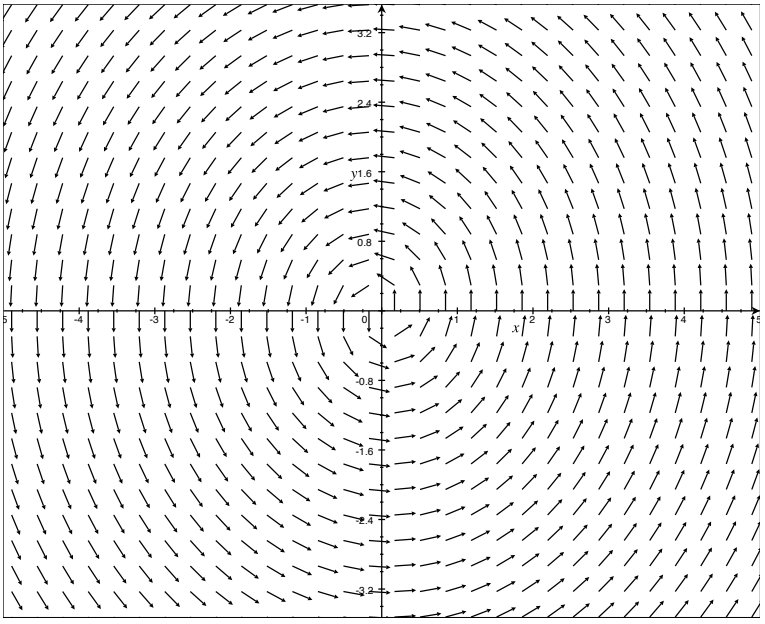
MAST20009
Vector Calculus
Problem Sheet Booklet



STUDENT NAME:

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For use of students of the University of Melbourne enrolled in the second year subject MAST20009 Vector Calculus.

Table of Contents

- Useful Formulae iv
- Vector Calculus Identities v
- Curvilinear Coordinates Formulae vi
- Problem Sheets 1
- Answers to Problem Sheets 27

Problem Sheets

There are seven problem sheets corresponding to the seven main topics covered in lectures. The problems can be grouped into two types:

- Questions labelled **Revision** cover material that is assumed knowledge from previous mathematics subjects. You should be able to complete all of these questions.
- Questions that do not have a label **Revision** are the core questions for MAST20009. It is essential that you attempt **all** of these problems.

Answers are at the back of the problem sheet booklet. Worked solutions will only be provided for selected questions.

Reference Book

There are no prescribed textbooks. The textbook recommended for extra reading on background material, proofs of theorems and applications is:

- **Vector Calculus**, Sixth Edition, by J. E. Marsden and A. J. Tromba, Freeman.

The book is available in the ERC Library. Any earlier or later edition of the book is also suitable.

This table gives references to the relevant section(s) of **Vector Calculus**, fourth edition, by Marsden and Tromba and to corresponding questions on the problem sheets.

Topic	**Textbook**	**Questions**
Functions of Several Variables		
Limits and continuity	§2.2	1-4
Partial differentiation revision	§2.3, §3.1	5-6
Differentiability	§2.3	7-9
Matrix version of chain rule	§2.5	10-11
Polar, cylindrical and spherical coordinates	§1.4	12-13, 38-41
Jacobian	pg 359	14
Taylor polynomials	§3.2	15-18
Extrema, constrained extrema	§3.3	19-20
Lagrange multipliers	§3.4	21-24
Space Curves and Vector Fields		
Vectors revision	§1.1-§1.3	25-27
Parametric paths - velocity, acceleration	§2.4, §4.1	28-31
Tangent vectors, curvature, torsion	pg 263-264	32-34
Parametrisation of surfaces	§7.3	35-37, 41
Tangent planes to parametrised surfaces	§7.3	36-37
Vector fields	§4.3	42
Flow lines	§4.3	43-44
Differential operators	§4.4	45-48, 51-55
Scalar and vector potentials	pg 497-500	49-50
Basic identities of vector analysis	§4.4	51-55

Topic	Textbook	Questions
Integrals over Paths		
Arc length	§4.2	56
Parametrisation of paths	§7.3	57
Path integrals	§7.1	58-59
Line integrals	§7.2	60-62
Double and Triple Integrals		
Double integrals	§5.1-§5.3	64-67
Areas and volumes using double integrals	§5.3	67
Change of order of integration	§5.4	68-69
Triple integrals	§5.6	70
Sketching surfaces revision	§2.1	71
Elementary regions	§5.6	72
Volumes using triple integrals	§5.6	73, 80
Change of variables for multiple integrals	§6.2	74-84
Averages, centre of mass, moment of inertia	§6.3	81-74
Integrals over Surfaces		
Area of a surface	§7.4	85
Integrals of scalar functions over surfaces	§7.5	86-87
Integrals of vector functions over surfaces	§7.6	88-89
Integral Theorems		
Green's theorem	§8.1	90-93
Divergence theorem in plane	§8.1	94-96
Stokes' theorem	§8.2	97-99
Conservative fields	§8.3	100-101
Gauss' divergence theorem	§8.4	102-104
Mixed integral theorems	§8.1-§8.4	105-106
Applications to physics and engineering	§8.5	107
General Curvilinear Coordinates		
Orthogonal curvilinear coordinates	None	108-111
Differential operators	None	108-111

Useful Formulae

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C$$

$$\int \sec x\,dx = \log|\sec x + \tan x| + C \qquad \int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + C$$

$$\int \sec^2 x\,dx = \tan x + C \qquad \int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

$$\int \sinh x\,dx = \cosh x + C \qquad \int \cosh x\,dx = \sinh x + C$$

$$\int \operatorname{sech}^2 x\,dx = \tanh x + C \qquad \int \operatorname{cosech}^2 x\,dx = -\coth x + C$$

$$\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \arcsin\left(\frac{x}{a}\right) + C \qquad \int \frac{1}{\sqrt{x^2 + a^2}}\,dx = \operatorname{arcsinh}\left(\frac{x}{a}\right) + C$$

$$\int \frac{-1}{\sqrt{a^2 - x^2}}\,dx = \arccos\left(\frac{x}{a}\right) + C \qquad \int \frac{1}{\sqrt{x^2 - a^2}}\,dx = \operatorname{arccosh}\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{a^2 + x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C \qquad \int \frac{1}{a^2 - x^2}\,dx = \frac{1}{a}\operatorname{arctanh}\left(\frac{x}{a}\right) + C$$

where $a > 0$ is constant and C is an arbitrary constant of integration.

$$\cos^2 x + \sin^2 x = 1 \qquad \cosh^2 x - \sinh^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x \qquad 1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\cot^2 x + 1 = \operatorname{cosec}^2 x \qquad \coth^2 x - 1 = \operatorname{cosech}^2 x$$

$$\cos(2x) = \cos^2 x - \sin^2 x \qquad \cosh(2x) = \cosh^2 x + \sinh^2 x$$

$$\cos(2x) = 2\cos^2 x - 1 \qquad \cosh(2x) = 2\cosh^2 x - 1$$

$$\cos(2x) = 1 - 2\sin^2 x \qquad \cosh(2x) = 1 + 2\sinh^2 x$$

$$\sin(2x) = 2\sin x\cos x \qquad \sinh(2x) = 2\sinh x\cosh x$$

$$\cos(x+y) = \cos x\cos y - \sin x\sin y \qquad \cosh(x+y) = \cosh x\cosh y + \sinh x\sinh y$$

$$\sin(x+y) = \sin x\cos y + \cos x\sin y \qquad \sinh(x+y) = \sinh x\cosh y + \cosh x\sinh y$$

$$\cosh x = \tfrac{1}{2}\left(e^x + e^{-x}\right) \qquad \sinh x = \tfrac{1}{2}\left(e^x - e^{-x}\right)$$

$$\cos x = \tfrac{1}{2}\left(e^{ix} + e^{-ix}\right) \qquad \sin x = \tfrac{1}{2i}\left(e^{ix} - e^{-ix}\right)$$

$$e^{ix} = \cos x + i\sin x \qquad \operatorname{arcsinh} x = \log(x + \sqrt{x^2+1})$$

$$\operatorname{arccosh} x = \log(x + \sqrt{x^2-1}) \qquad \operatorname{arctanh} x = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

Basic Identities of Vector Calculus

Let f and $g : \mathbb{R}^3 \to \mathbb{R}$ be scalar functions, $\boldsymbol{F}$ and $\boldsymbol{G} : \mathbb{R}^3 \to \mathbb{R}^3$ be vector fields, and $\beta \in \mathbb{R}$ be any constant.

1. $\boldsymbol{\nabla}(f+g) = \boldsymbol{\nabla} f + \boldsymbol{\nabla} g$
2. $\boldsymbol{\nabla}(\beta f) = \beta \boldsymbol{\nabla} f$
3. $\boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla} g + g\boldsymbol{\nabla} f$
4. $\boldsymbol{\nabla}\left(\dfrac{f}{g}\right) = \dfrac{g\boldsymbol{\nabla} f - f\boldsymbol{\nabla} g}{g^2}$ provided $g \neq 0$.
5. $\boldsymbol{\nabla} \cdot (\boldsymbol{F} + \boldsymbol{G}) = \boldsymbol{\nabla} \cdot \boldsymbol{F} + \boldsymbol{\nabla} \cdot \boldsymbol{G}$
6. $\boldsymbol{\nabla} \times (\boldsymbol{F} + \boldsymbol{G}) = \boldsymbol{\nabla} \times \boldsymbol{F} + \boldsymbol{\nabla} \times \boldsymbol{G}$
7. $\boldsymbol{\nabla} \cdot (f\boldsymbol{F}) = f\boldsymbol{\nabla} \cdot \boldsymbol{F} + \boldsymbol{F} \cdot \boldsymbol{\nabla} f$
8. $\boldsymbol{\nabla} \cdot (\boldsymbol{F} \times \boldsymbol{G}) = \boldsymbol{G} \cdot (\boldsymbol{\nabla} \times \boldsymbol{F}) - \boldsymbol{F} \cdot (\boldsymbol{\nabla} \times \boldsymbol{G})$
9. $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{F}) = 0$
10. $\boldsymbol{\nabla} \times (f\boldsymbol{F}) = f\boldsymbol{\nabla} \times \boldsymbol{F} + \boldsymbol{\nabla} f \times \boldsymbol{F}$
11. $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \boldsymbol{0}$
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2\boldsymbol{\nabla} f \cdot \boldsymbol{\nabla} g$
13. $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} f \times \boldsymbol{\nabla} g) = 0$
14. $\boldsymbol{\nabla} \cdot (f\boldsymbol{\nabla} g - g\boldsymbol{\nabla} f) = f\nabla^2 g - g\nabla^2 f$
15. $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{F}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{F}) - \nabla^2 \boldsymbol{F}$

Note:
The identities require $f, g, \boldsymbol{F}$ and $\boldsymbol{G}$ to be suitably differentiable, either order C^1 or C^2.

Grad, Div, Curl, and Laplacian in Orthogonal Curvilinear Coordinates

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a C^2 scalar function and $\boldsymbol{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a C^1 vector field where

$$\boldsymbol{F}(u_1, u_2, u_3) = F_1(u_1, u_2, u_3)\boldsymbol{e}_1 + F_2(u_1, u_2, u_3)\boldsymbol{e}_2 + F_3(u_1, u_2, u_3)\boldsymbol{e}_3.$$

Then

1. $\boldsymbol{\nabla} f = \dfrac{1}{h_1}\dfrac{\partial f}{\partial u_1}\boldsymbol{e}_1 + \dfrac{1}{h_2}\dfrac{\partial f}{\partial u_2}\boldsymbol{e}_2 + \dfrac{1}{h_3}\dfrac{\partial f}{\partial u_3}\mathbf{e_3}$

2. $\boldsymbol{\nabla} \cdot \boldsymbol{F} = \dfrac{1}{h_1 h_2 h_3}\left[\dfrac{\partial\left(h_2 h_3 F_1\right)}{\partial u_1} + \dfrac{\partial\left(h_1 h_3 F_2\right)}{\partial u_2} + \dfrac{\partial\left(h_1 h_2 F_3\right)}{\partial u_3}\right]$

3. $\boldsymbol{\nabla} \times \boldsymbol{F} = \dfrac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\boldsymbol{e}_1 & h_2\boldsymbol{e}_2 & h_3\boldsymbol{e}_3 \\ \dfrac{\partial}{\partial u_1} & \dfrac{\partial}{\partial u_2} & \dfrac{\partial}{\partial u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}$

4. $\nabla^2 f = \dfrac{1}{h_1 h_2 h_3}\left[\dfrac{\partial}{\partial u_1}\left(\dfrac{h_2 h_3}{h_1}\dfrac{\partial f}{\partial u_1}\right) + \dfrac{\partial}{\partial u_2}\left(\dfrac{h_1 h_3}{h_2}\dfrac{\partial f}{\partial u_2}\right) + \dfrac{\partial}{\partial u_3}\left(\dfrac{h_1 h_2}{h_3}\dfrac{\partial f}{\partial u_3}\right)\right]$

Note: Equations 1-4 reduce to the usual expressions for cartesian coordinates if

$$h_1 = h_2 = h_3 = 1; \quad \boldsymbol{e}_1 = \boldsymbol{i}, \boldsymbol{e}_2 = \boldsymbol{j}, \boldsymbol{e}_3 = \boldsymbol{k}; \quad (u_1, u_2, u_3) = (x, y, z).$$

Cylindrical Coordinates

Cylindrical coordinates (ρ, ϕ, z) are defined by

$$x = \rho\cos\phi, \quad y = \rho\sin\phi, \quad z = z$$

where $\rho \geq 0$, $0 \leq \phi \leq 2\pi$. Then $(u_1, u_2, u_3) = (\rho, \phi, z)$ and $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$.

Equations 1-4 reduce to:

1. $\boldsymbol{\nabla} f = \dfrac{\partial f}{\partial \rho}\,\hat{\boldsymbol{\rho}} + \dfrac{1}{\rho}\dfrac{\partial f}{\partial \phi}\,\hat{\boldsymbol{\phi}} + \dfrac{\partial f}{\partial z}\,\hat{\boldsymbol{z}}$

2. $\boldsymbol{\nabla}\cdot\boldsymbol{F} = \dfrac{1}{\rho}\left[\dfrac{\partial(\rho F_1)}{\partial \rho} + \dfrac{\partial F_2}{\partial \phi} + \dfrac{\partial(\rho F_3)}{\partial z}\right] = \dfrac{1}{\rho}\left(F_1 + \rho\dfrac{\partial F_1}{\partial \rho}\right) + \dfrac{1}{\rho}\dfrac{\partial F_2}{\partial \phi} + \dfrac{\partial F_3}{\partial z}$

3. $$\boldsymbol{\nabla}\times\boldsymbol{F} = \frac{1}{\rho}\begin{vmatrix} \hat{\boldsymbol{\rho}} & \rho\hat{\boldsymbol{\phi}} & \hat{\boldsymbol{z}} \\ \dfrac{\partial}{\partial \rho} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ F_1 & \rho F_2 & F_3 \end{vmatrix}$$

$$= \frac{1}{\rho}\left[\left(\frac{\partial F_3}{\partial \phi} - \frac{\partial(\rho F_2)}{\partial z}\right)\hat{\boldsymbol{\rho}} - \left(\frac{\partial F_3}{\partial \rho} - \frac{\partial F_1}{\partial z}\right)\rho\hat{\boldsymbol{\phi}} + \left(\frac{\partial(\rho F_2)}{\partial \rho} - \frac{\partial F_1}{\partial \phi}\right)\hat{\boldsymbol{z}}\right]$$

4. $\nabla^2 f = \dfrac{1}{\rho}\left[\dfrac{\partial}{\partial \rho}\left(\rho\dfrac{\partial f}{\partial \rho}\right) + \dfrac{\partial}{\partial \phi}\left(\dfrac{1}{\rho}\dfrac{\partial f}{\partial \phi}\right) + \dfrac{\partial}{\partial z}\left(\rho\dfrac{\partial f}{\partial z}\right)\right] = \dfrac{1}{\rho}\dfrac{\partial f}{\partial \rho} + \dfrac{\partial^2 f}{\partial \rho^2} + \dfrac{1}{\rho^2}\dfrac{\partial^2 f}{\partial \phi^2} + \dfrac{\partial^2 f}{\partial z^2}$

Spherical Coordinates

Spherical coordinates (r, θ, ϕ) are defined by

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

where $r \geq 0$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$. Then $(u_1, u_2, u_3) = (r, \theta, \phi)$ and $h_1 = 1$, $h_2 = r$, $h_3 = r\sin\theta$.

Equations 1-4 reduce to:

1. $\boldsymbol{\nabla} f = \dfrac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \dfrac{1}{r}\dfrac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \dfrac{1}{r\sin\theta}\dfrac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}}$

2. $$\boldsymbol{\nabla}\cdot\boldsymbol{F} = \frac{1}{r^2\sin\theta}\left[\frac{\partial\left(r^2\sin\theta F_1\right)}{\partial r} + \frac{\partial\left(r\sin\theta F_2\right)}{\partial\theta} + \frac{\partial\left(rF_3\right)}{\partial\phi}\right]$$
$$= \frac{1}{r^2}\frac{\partial\left(r^2F_1\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial\left(\sin\theta F_2\right)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial F_3}{\partial\phi}$$

3. $$\boldsymbol{\nabla}\times\boldsymbol{F} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{\boldsymbol{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\hat{\boldsymbol{\phi}} \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial\phi} \\ F_1 & rF_2 & r\sin\theta F_3 \end{vmatrix}$$
$$= \frac{1}{r^2\sin\theta}\left[\left(\frac{\partial\left(r\sin\theta F_3\right)}{\partial\theta} - \frac{\partial\left(rF_2\right)}{\partial\phi}\right)\hat{\boldsymbol{r}} - \left(\frac{\partial\left(r\sin\theta F_3\right)}{\partial r} - \frac{\partial F_1}{\partial\phi}\right)r\hat{\boldsymbol{\theta}} + \left(\frac{\partial\left(rF_2\right)}{\partial r} - \frac{\partial F_1}{\partial\theta}\right)r\sin\theta\hat{\boldsymbol{\phi}}\right]$$

4. $$\nabla^2 f = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial f}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{\partial}{\partial\phi}\left(\frac{1}{\sin\theta}\frac{\partial f}{\partial\phi}\right)\right]$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$$

MAST20009 Vector Calculus
Problem Sheet 1
Functions of Several Variables

1. Limits. Evaluate the following limits, if they exist.

(a) $\displaystyle\lim_{(x,y)\to(1,0)} (x^3 + y^4 + 3)$;

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2 + 2}$;

(c) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3y^2}{2x^2 - y^2}$;

(d) $\displaystyle\lim_{(x,y)\to(2,1)} \frac{2x^4 + 3y^3}{3x^2 - 5}$;

(e) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4 + y^4}$;

(f) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{(x-y)^2}{x^2 + y^2}$;

(g) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^4 - y^4}{x^2 + y^2}$;

(h) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{y - 2x + \sin 2x}{x^3 + y}$;

(i) $\displaystyle\lim_{(x,y)\to(1,-1)} \frac{x + y}{2x^2 + 3xy + y^2}$;

(j) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x}{x^2 + y^2}$.

2. No Limits. Show that the following limits do not exist.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{(x^2 + y)^3}$;

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{2x^4y^4}{(3x^4 + y^2)^3}$.

Hint: You may need to examine the limit along $y = x^2$ as well as the usual lines.

3. Limit Exists. Using the Sandwich Theorem, show that

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{7x^2y^2}{x^2 + 2y^4} = 0$;

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3yx^2}{x^2 + y^2} = 0$.

4. Continuity. Consider the following functions f where

(i) $f(x, y) = ye^x + \sin x$;

(ii) $f(x, y) = \dfrac{\sin(x^2 + y^2)}{x^2 + y^2}$;

(iii) $f(x, y) = \dfrac{xy}{x^2 + y^2}$.

(a) Which functions are continuous at $(x, y) = (0, 0)$?

(b) Can the functions which are not continuous at $(0, 0)$, be made continuous at $(0, 0)$ by suitably defining them at the origin?

5. Partial Derivatives Revision. For each function f, find expressions for the partial derivatives listed.

(a) $f(x, y) = \log(x + 3y)$. Find $f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$.

(b) $f(x, y) = e^{xy}$. Find $f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy}$.

(c) $f(x, y, z) = x^2y^3z^5$. Find $f_x, f_y, f_z, f_{yy}, f_{yz}, f_{xyz}$.

(d) $f(x, y, z) = \dfrac{x}{y} + \dfrac{y}{z}$. Find $f_x, f_y, f_z, f_{yz}, f_{yy}, f_{xyy}$.

6. Laplace's Equation. Let

$$V(x, y, z) = -\frac{GmM}{\sqrt{x^2 + y^2 + z^2}}$$

where $(x, y, z) \neq (0, 0, 0)$ and G, m, M are constants. Show that V satisfies Laplace's equation, that is,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0.$$

7. Differentiable Functions. Find the largest possible domain for each function f. Show that the function is C^1 (and is therefore differentiable) in this domain.

(a) $f(x, y) = \dfrac{x}{y} + \dfrac{y}{x}$; (b) $f(x, y) = \dfrac{xy}{\sqrt{x^2 + y^2}}$; (c) $f(r, \theta) = \dfrac{1}{2} r \sin 2\theta, \quad r > 0$.

8. Continuity and Differentiability. Consider the following function f given by

$$f(x, y) = \begin{cases} \dfrac{4x^2 - y^3}{x^2 + 5y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

(a) Calculate $\dfrac{\partial f}{\partial y}$ if $(x, y) \neq (0, 0)$.

(b) Calculate $\dfrac{\partial f}{\partial y}$ if $(x, y) = (0, 0)$.

(c) Is $\dfrac{\partial f}{\partial y}$ continuous at $(0, 0)$? Explain.

(d) Is the function C^1 at $(0, 0)$?

9. Different Mixed Partials. Let $f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$.

(a) If $(x,y) \neq (0,0)$, calculate $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

(b) Show that $\dfrac{\partial f}{\partial x} = 0$ and $\dfrac{\partial f}{\partial y} = 0$ at $(x,y) = (0,0)$.

(c) Show that $\dfrac{\partial^2 f}{\partial x\, \partial y} = 1$ and $\dfrac{\partial^2 f}{\partial y\, \partial x} = -1$ at $(x,y) = (0,0)$. Why are the mixed partials different?

10. Matrix Chain Rule. For each of the following, find the specified derivative using the matrix version of the chain rule. In each case, check your answer by direct substitution.

(a) Let $p(t) = (t, t^2, t^3)$ and $f(x,y,z) = 3x^2 + 4xy + z^2$. Evaluate $\boldsymbol{D}(f \circ p)$ at $t = 1$.

(b) Let $f(x,y) = x^2 - y^2 + x^3 y$ and $(x,y) = (r\cos\theta, r\sin\theta)$. Find the derivative of

$f[x(r,\theta), y(r,\theta)]$ at $(r,\theta) = \left(2, \dfrac{\pi}{4}\right)$.

(c) Let $f(u,v,w) = (u - v, uvw)$ and $g(x,y) = (xy, x^2, y^2)$. Evaluate $\boldsymbol{D}(f \circ g)$ at $(2,1)$.

11. Matrix Chain Rule Again. Using the matrix chain rule, evaluate the derivative of

(a) $f(g[f(x,y)])$ at $(0,1)$ where $f(x,y) = (x^2, 2y, x-y)$ and $g(x,y,z) = (x+z, y^2)$.

(b) $g\,[f(g[f(x,y)])]$ at $(1,0)$ where $f(x,y) = (x^2, x^2-y^2, x^2+y^2)$, $g(x,y,z) = (x^2-z^2, y^2+z^2)$.

12. Cylindrical Coordinates. The relation between three-dimensional Cartesian coordinates (X,Y,Z) and cylindrical coordinates (ρ,ϕ,z) is given by

$$X = \rho\cos\phi, \qquad Y = \rho\sin\phi, \qquad Z = z,$$

where $\rho > 0$, $0 \leq \phi < 2\pi$ and $z \in \mathbb{R}$.

(a) Find the Jacobi matrix of this transformation, and hence find its Jacobian.

(b) Where possible solve these equations for ρ, ϕ and z as functions of X, Y, Z.

(c) Calculate the partial derivatives $\dfrac{\partial \rho}{\partial X}$, $\dfrac{\partial \phi}{\partial Y}$, and $\dfrac{\partial z}{\partial Z}$ in terms of ρ, ϕ, z.

13. Spherical Coordinates. The relation between three-dimensional Cartesian coordinates (x, y, z) and spherical coordinates (r, θ, ϕ) is given by

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta,$$

where $r > 0$, $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$.

(a) Find the Jacobi matrix of this transformation, and hence find its Jacobian.

(b) Where possible solve these equations for r, θ, and ϕ as functions of x, y, z.

(c) Calculate the partial derivatives $\dfrac{\partial r}{\partial x}$, $\dfrac{\partial \theta}{\partial y}$ and $\dfrac{\partial \phi}{\partial z}$ in terms of r, θ, ϕ.

14. Product of Jacobians. Assume that $u = u(r, s)$ and $v = v(r, s)$ are C^1 functions of r and s. Also, assume that $r = r(x, y)$ and $s = s(x, y)$ are C^1 functions of x and y. Prove that

$$\frac{\partial(u, v)}{\partial(r, s)}\,\frac{\partial(r, s)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(x, y)}.$$

(Hint: Use the chain rule and the identity $\det(AB) = \det(A)\det(B)$.)

15. Taylor Polynomials. Determine the Taylor polynomial of order n for each function f about the given point (x_0, y_0).

(a) $f(x, y) = e^{(x+y)}, \quad n = 3, \quad (x_0, y_0) = (0, 0);$

(b) $f(x, y) = \sin(xy) + \cos(xy), \quad n = 2, \quad (x_0, y_0) = (0, 0);$

(c) $f(x, y) = \dfrac{1}{1 + x - y}, \quad n = 2, \quad (x_0, y_0) = (1, 1);$

(d) $f(x, y) = \cos(4x - 3y), \quad n = 3, \quad (x_0, y_0) = \left(\dfrac{\pi}{8}, 0\right).$

16. Taylor Polynomial Application. Let $f(x, y) = e^{(x-1)^2}\cos y$.

(a) Determine the Taylor polynomial of degree 2 for f about the point $(1, 0)$.

(b) Using your answer to part (a), approximate $f(1.1, 0.2)$. Compare your approximate answer to the exact value.

(c) Using your answer to part (a), approximate the double integral:

$$\int_0^1 \int_0^1 f(x, y)\,dx\,dy.$$

17. Taylor Remainder. For a certain C^2 function $f(x, y)$ it is known that

$$f(0,0) = 1, \ f_x(0,0) = 0.25, \ f_y(0,0) = 0.50$$

and that

$$|f_{xx}| \leq 0.15, \ |f_{xy}| \leq 0.05, \ |f_{yy}| \leq 0.05$$

everywhere along that segment of the line $y = x$ which joins $(0, 0)$ and $(0.1, 0.1)$.

(a) Determine the Taylor polynomial of degree 1 for f about the point $(0, 0)$.

(b) Using your answer to part (a), approximate $(0.1, 0.1)$ and determine the error R_1 in your approximation.

(c) Show that

$$1.0735 \leq f(0.1, 0.1) \leq 1.0765.$$

18. Another Taylor Remainder. Consider the function

$$g(x, y) = e^{3-2x+y}.$$

(a) Determine the second order Taylor polynomial for g about the point $(1, -1)$. Hence, approximate $g(0.9, -1.1)$.

(b) Determine an upper bound for the error in your approximation of $g(0.9, -1.1)$ using the remainder formula.

19. Maxima and Minima Revision. Find all the local minima, local maxima and saddle points of the following functions.

(a) $f(x, y) = e^{(1+x^2-y^2)}$;

(b) $f(x, y) = x^3 - 12xy + 8y^3$;

(c) $f(x, y) = x^2 + y^2 + 2xy$.

20. Eliminate Variable. To find the critical points of the function of three variables

$$f(x, y, z) = x^2 - y^2 + yz$$

subject to the condition $z = y^2$, it is possible to eliminate one variable.

(a) Eliminate the variable z and rewrite f as a function of two variables $F(x, y)$.

(b) Find and classify all of the critical points of F.

(c) Hence find and classify all of the critical points of f.

21. Use Lagrange Multipliers. Redo question 20, using Lagrange multipliers to find the critical points of

$$f(x, y, z) = x^2 - y^2 + yz$$

subject to the condition $z = y^2$.

22. Lagrange Multipliers. Find the extrema of the following functions subject to the given constraints.

(a) $f(x, y, z) = x - y + z, \qquad x^2 + y^2 + z^2 = 2;$

(b) $f(x, y) = x, \qquad x^2 + 2y^2 = 3;$

(c) $f(x, y, z) = xy + yz, \qquad xz = 1;$

(d) $f(x, y, z) = x + y + z, \qquad 2x + z = 1, \qquad 2x^2 + y^2 = 1;$

(e) $f(x, y, z) = x, \qquad x^2 + 2y^2 + 2z^2 = 8, \qquad x + y = z.$

23. More Lagrange Multipliers. (a) Design a cylindrical can with a lid to contain 1 litre (1000 cm^3) of water, using the minimum amount of metal.

(b) Determine the points on the ellipse defined as the intersection of the surfaces $x + y = 1$ and $x^2 + 2y^2 + z^2 = 1$ which are nearest to the origin. (Hint: Minimise the square of the distance from a point on the ellipse to the origin.)

24. Inequality Constraints. (a) Find the absolute maxima and minima of the function f given by $f(x, y) = 5x^2 - 2y^2 + 10$ on the disk $x^2 + y^2 \leq 1$.

(b) A parcel delivery service requires that the dimension of a rectangular box be such that the length plus twice the width plus twice the height be no more than 108 centimetres. What is the volume of the largest box that the company will deliver?

MAST20009 Vector Calculus

Problem Sheet 2

Space Curves and Vector Fields

25. Vectors Revision. (a) Evaluate the dot and cross products

$$\text{(i)} \quad (2\boldsymbol{i}+\boldsymbol{j})\cdot(-\boldsymbol{j}+\boldsymbol{k}); \qquad \text{(ii)} \quad (2\boldsymbol{i}+\boldsymbol{j})\times(-\boldsymbol{j}+\boldsymbol{k}).$$

(b) Find a unit vector perpendicular to $\boldsymbol{i}+\boldsymbol{j}+\boldsymbol{k}$ and $2\boldsymbol{i}-3\boldsymbol{j}+4\boldsymbol{k}$.

(c) What restrictions must be made on b so that $2\boldsymbol{i}+b\boldsymbol{j}$ is orthogonal to $-3\boldsymbol{i}+2\boldsymbol{j}+\boldsymbol{k}$?

26. Equation of Lines Revision. Find the equations of the following lines in both vector and parametric form.

(a) The line passing through the point $(0,1,2)$ and parallel to $\boldsymbol{i}-2\boldsymbol{j}+\boldsymbol{k}$;

(b) The line passing through the points $(1,2,1)$ and $(4,-2,3)$.

27. Equation of Planes Revision. Find the Cartesian equations of the following planes.

(a) The plane passing through the points $(1,1,1)$, $(3,0,2)$, and $(4,3,1)$;

(b) The plane passing through the point $(1,0,2)$ and perpendicular to the vector $\boldsymbol{i}+2\boldsymbol{j}-\boldsymbol{k}$.

28. Parametric curves. For each parametric curve $\boldsymbol{c}$, find the velocity, acceleration, and the equation of the tangent line at the given value of t.

$$\text{(a) } \boldsymbol{c}(t)=\cos t\boldsymbol{i}+\sin(2t)\boldsymbol{j}, \quad t=0; \qquad \text{(b) } \boldsymbol{c}(t)=\sqrt{2}t\boldsymbol{i}+e^{t}\boldsymbol{j}+e^{-t}\boldsymbol{k}, \quad t=0.$$

29. Rolling Wheel. A wheel of radius R rolls to the right along a straight line at speed v. The path of a point on the rim is given by

$$\boldsymbol{c}(t)=\left(vt-R\sin\left(\frac{vt}{R}\right), R-R\cos\left(\frac{vt}{R}\right)\right).$$

When is the velocity vector of this particular point horizontal? What is the speed at this point?

30. Velocity Properties. Suppose the path of a moving particle is given by $\boldsymbol{c}(t)$. Let $\boldsymbol{v}(t)$ be its velocity and $\boldsymbol{a}(t)$ the acceleration at time t. Suppose that $\boldsymbol{F}$ is a C^1 vector field, $m>0$, and $\boldsymbol{F}[\boldsymbol{c}(t)]=m\boldsymbol{a}(t)$. Prove that the rate of change of angular momentum equals the torque, that is,

$$\frac{d}{dt}\left[m\boldsymbol{c}(t)\times\boldsymbol{v}(t)\right]=\boldsymbol{c}(t)\times\boldsymbol{F}\left[\boldsymbol{c}(t)\right].$$

31. Differentiation. (a) If $\boldsymbol{F}$ is a differentiable function of t, find the derivative of

$$\boldsymbol{F} \cdot \left(\frac{d\boldsymbol{F}}{dt} \times \frac{d^2\boldsymbol{F}}{dt^2} \right).$$

(b) Suppose the function $\boldsymbol{u} : \mathbb{R} \to \mathbb{R}^3$ is such that $\boldsymbol{u}'''$ exists. Evaluate and simplify

$$\frac{d}{dt}\left[(\boldsymbol{u} + \boldsymbol{u}'') \cdot (\boldsymbol{u} \times \boldsymbol{u}')\right].$$

32. Curve Properties. Consider the curve given by

$$\boldsymbol{c}(t) = (\sin t - t\cos t, \cos t + t\sin t, 27); \qquad t > 0.$$

(a) Find the unit tangent, unit normal and unit binormal vectors to the curve.

(b) Compute the curvature and torsion for the curve.

33. More curve Properties. Consider the curve given by

$$\mathbf{c}(t) = (e^{2t}\sin t, e^{2t}\cos t, 1).$$

(a) Find the unit tangent, unit normal, and unit binormal vectors to the curve.

(b) Compute the curvature and torsion for the curve.

34. Curve Properties. Let $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ be the unit tangent, normal and binormal vectors to a C^3 curve. Prove that

(a) $\dfrac{d\boldsymbol{B}}{dt} \cdot \boldsymbol{B} = 0$; (b) $\dfrac{d\boldsymbol{B}}{dt} \cdot \boldsymbol{T} = 0$; (c) $\dfrac{d\boldsymbol{B}}{dt}$ is a scalar multiple of $\boldsymbol{N}$.

35. Parametrised Normals. Find a unit vector normal to the following parametrised surfaces in terms of x, y and z. Identify each surface.

(a) $x = \cos v \sin u, \quad y = \sin v \sin u, \quad z = \cos u \quad$ where $0 \le u \le \pi, \quad 0 \le v \le 2\pi$.

(b) $x = \sin v, \quad y = u, \quad z = \cos v \quad$ where $-1 \le u \le 3, \quad 0 \le v \le 2\pi$.

36. Parametric Surface 1. Consider the parametric surface defined by:

$$x = 2u, \quad y = u^2 + v, \quad z = v^2.$$

(a) Find a vector normal to the surface in terms of u and v.

(b) For what values of u and v is the surface smooth?

(c) Find the equation of the tangent plane to the surface at $(0, 1, 1)$.

37. Parametric Surface 2. Consider the parametric surface defined by:

$$x = u^2 - v^2, \quad y = u + v, \quad z = u^2 + 4v.$$

(a) Find a vector normal to the surface in terms of u and v.

(b) For what values of u and v is the surface smooth?

(c) Find the equation of the tangent plane to the surface at $\left(-\frac{1}{4}, \frac{1}{2}, 2\right)$.

38. Convert from Polar. (a) The following points are given in polar coordinates (r, θ). Express each point in cartesian coordinates (x, y).

$$\text{(i)} \left(7, \frac{2\pi}{3}\right); \qquad \text{(ii)} \left(6, \frac{-5\pi}{6}\right); \qquad \text{(iii)} \ (0, -\pi).$$

(b) The following points are given in cartesian coordinates (x, y). Express each point in polar coordinates (r, θ).

$$\text{(i)} \left(2\sqrt{3}, -2\right); \qquad \text{(ii)} \ (0, -2); \qquad \text{(iii)} \ (-8, -8).$$

39. Convert from Cylindrical. The following points are given in cylindrical coordinates (ρ, ϕ, z). Express each point in cartesian coordinates (x, y, z) and spherical coordinates (r, θ, ϕ).

$$\text{(a)} \ (1, \frac{\pi}{4}, 1); \qquad \text{(b)} \left(2, \frac{\pi}{2}, -4\right); \qquad \text{(c)} \left(3, \frac{\pi}{6}, 4\right); \qquad \text{(d)} \left(2, \frac{3\pi}{4}, -2\right).$$

40. Convert from Cartesian. The following points are given in cartesian coordinates (x, y, z). Express each point in spherical coordinates (r, θ, ϕ) and cylindrical coordinates (ρ, ϕ, z).

$$\text{(a)} \ (2, 1, -2); \qquad \text{(b)} \ (0, 3, 4); \qquad \text{(c)} \ (\sqrt{2}, 1, 1); \qquad \text{(d)} \ (-2\sqrt{3}, -2, 3).$$

41. Parametric Surfaces. Find a parametrisation for the following surfaces based on either cylindrical or spherical coordinates, as appropriate.

(a) The cylinder $x^2 + z^2 = 2, \ 0 \leq y \leq 2$.

(b) The cone $z = 15 - 3\sqrt{x^2 + y^2}$ for $z \geq 0$.

(c) The hemisphere $z = -\sqrt{9 - x^2 - y^2}$.

42. Vector Fields. Sketch the following vector fields $\boldsymbol{F}$ where

$$\text{(a)} \ \boldsymbol{F}(x, y) = (2, 2); \qquad \text{(b)} \ \boldsymbol{F}(x, y) = (x, y); \qquad \text{(b)} \ \boldsymbol{F}(x, y) = (2y, x).$$

43. Flow Lines. Sketch a few flow lines of the following vector fields $\boldsymbol{F}$. Derive the differential equations for the flow lines and solve them to obtain an expression for the flow lines.

$$\text{(a)} \ \boldsymbol{F}(x, y) = (y, -x); \qquad \text{(b)} \ \boldsymbol{F}(x, y) = (x, -y); \qquad \text{(c)} \ \boldsymbol{F}(x, y, z) = (y, -x, 0).$$

44. Flow Line Properties. Prove that the curve given by $\boldsymbol{c}(t)$ is a flow line of the given velocity vector field $\boldsymbol{F}$.

$$\text{(a) } \boldsymbol{c}(t) = \left(e^{2t}, \log|t|, \frac{1}{t}\right), \quad t \neq 0, \quad \boldsymbol{F}(x,y,z) = (2x, z, -z^2);$$

$$\text{(b) } \boldsymbol{c}(t) = (\sin t, \cos t, e^t), \quad \boldsymbol{F}(x,y,z) = (y, -x, z).$$

45. Divergence. Find the divergence of the following vector fields $\boldsymbol{V}$.

(a) $\boldsymbol{V}(x,y,z) = yz\boldsymbol{i} + xz\boldsymbol{j} + xy\boldsymbol{k}$;

(b) $\boldsymbol{V}(x,y,z) = x\boldsymbol{i} + (y + \cos x)\boldsymbol{j} + (z + e^{xy})\boldsymbol{k}$;

(c) $\boldsymbol{V}(x,y) = \sin(xy)\boldsymbol{i} - \cos(x^2y)\boldsymbol{j}$.

46. Curl. Find the curl of the following vector fields $\boldsymbol{V}$.

(a) $\boldsymbol{V}(x,y,z) = xy\boldsymbol{i} - \sin z\boldsymbol{j} + \boldsymbol{k}$;

(b) $\boldsymbol{V}(x,y,z) = yz\boldsymbol{i} + xz\boldsymbol{j} + xy\boldsymbol{k}$;

(c) $\boldsymbol{V}(x,y,z) = (x^2 + y^2 + z^2)(3\boldsymbol{i} + 4\boldsymbol{j} + 5\boldsymbol{k})$;

47. Div, Grad, and Curl. Let $\boldsymbol{F}(x,y,z) = 2xz^2\boldsymbol{i} + \boldsymbol{j} + y^3zx\boldsymbol{k}$, $\quad f(x,y,z) = x^2y$. Find expressions for the following quantities where they are defined.

(a) $\boldsymbol{\nabla} f$;	(b) $\boldsymbol{\nabla} \times \boldsymbol{F}$;	(c) $\boldsymbol{\nabla} \cdot \boldsymbol{F}$;
(d) $\boldsymbol{\nabla} \times f$;	(e) $\boldsymbol{F} \times \boldsymbol{\nabla} f$;	(f) $\boldsymbol{F} \cdot \boldsymbol{\nabla} f$;
(g) $\boldsymbol{\nabla} \cdot f$;	(h) $\nabla^2 f$;	(i) $\nabla^2 \boldsymbol{F}$.

48. Curl of Grad. Show that $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \boldsymbol{0}$ for the following functions f.

(a) $f(x,y,z) = \sqrt{x^2 + y^2 + z^2}$; $\qquad$ (b) $f(x,y,z) = xy + yz + xz$.

49. Scalar Potentials. Consider the following vector fields $\boldsymbol{F}$. Which vector fields are irrotational? For each irrotational vector field, find a scalar function ϕ such that $\boldsymbol{F} = \boldsymbol{\nabla}\phi$.

(a) $\boldsymbol{F}(x,y,z) = 2x\boldsymbol{i} + 3y\boldsymbol{j} + 4z\boldsymbol{k}$;

(b) $\boldsymbol{F}(x,y,z) = y\cos x\boldsymbol{i} + x\sin y\boldsymbol{j}$;

(c) $\boldsymbol{F}(x,y,z) = 2xye^z\boldsymbol{i} + x^2e^z\boldsymbol{j} + (x^2ye^z + z^2)\boldsymbol{k}$.

50. Finding Vector Fields. Consider the following vector fields $\boldsymbol{F}$. Which vector fields are incompressible? For each incompressible vector field, find a vector $\boldsymbol{G}$ such that $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{G}$.

(a) $\boldsymbol{F}(x, y, z) = xz\boldsymbol{i} - yz\boldsymbol{j} + y\boldsymbol{k}$;

(b) $\boldsymbol{F}(x, y, z) = x\cos y\boldsymbol{i} - \sin y\boldsymbol{j} + \sin x\boldsymbol{k}$;

(c) $\boldsymbol{F}(x, y, z) = 2xye^z\boldsymbol{i} + x^2e^z\mathbf{j} + (x^2ye^z + z^2)\boldsymbol{k}$.

51. Vector Identity One. (a) Let f and g be C^1 scalar functions. Prove that

$$\boldsymbol{\nabla}\left(\frac{f}{g}\right) = \frac{1}{g^2}\left(g\boldsymbol{\nabla}f - f\boldsymbol{\nabla}g\right), \quad g \neq 0.$$

(b) Using (a), evaluate $\boldsymbol{\nabla}\left(\dfrac{e^{r^2}}{r}\right)$ where $\boldsymbol{r}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$ and $r = |\boldsymbol{r}|$.

52. Vector Identity Two. (a) Let $\boldsymbol{F}$ be a C^1 vector field and f be a C^1 scalar function. Prove that

$$\boldsymbol{\nabla} \cdot (f\boldsymbol{F}) = f\boldsymbol{\nabla} \cdot \boldsymbol{F} + \boldsymbol{F} \cdot \boldsymbol{\nabla}f.$$

(b) Using (a), evaluate $\boldsymbol{\nabla} \cdot \left(\dfrac{\boldsymbol{r}}{r^5}\right)$ where $\boldsymbol{r}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$ and $r = |\boldsymbol{r}|$.

53. Vector Identity Three. (a) Let $\boldsymbol{F}$ be a C^1 vector field and f be a C^1 scalar function. Prove that

$$\boldsymbol{\nabla} \times (f\boldsymbol{F}) = f\boldsymbol{\nabla} \times \boldsymbol{F} + \boldsymbol{\nabla}f \times \boldsymbol{F}.$$

(b) Using (a), evaluate $\boldsymbol{\nabla} \times \left(\dfrac{\boldsymbol{r}}{r^3}\right)$ where $\boldsymbol{r}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$ and $r = |\boldsymbol{r}|$.

54. Using Vector Identities. Let f, g, and h be any C^2 scalar functions. Using the standard identities of vector calculus, prove that

$$\boldsymbol{\nabla} \cdot (f\boldsymbol{\nabla}g \times \boldsymbol{\nabla}h) = \boldsymbol{\nabla}f \cdot (\boldsymbol{\nabla}g \times \boldsymbol{\nabla}h).$$

55. Vector Properties. Let $\boldsymbol{r}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$ and $r = |\boldsymbol{r}|$. Using the standard identities of vector calculus, or otherwise, prove that

(a) $\boldsymbol{\nabla}(r^n) = nr^{n-2}\boldsymbol{r}$;

(b) $\boldsymbol{\nabla} \cdot (r^n\boldsymbol{r}) = (n + 3)r^n$;

(c) $\nabla^2(r^n) = n(n + 1)r^{n-2}$;

(d) $\boldsymbol{\nabla} \times (r^n\boldsymbol{r}) = \mathbf{0}$;

where $n \in \mathbb{R}$.

MAST20009 Vector Calculus

Problem Sheet 3

Integrals over Paths

56. Arc Length. Find the arc length of the given curve on the specified interval.

(a) $\boldsymbol{c}(t) = (2\cos t, 2\sin t, t); \quad 0 \le t \le 2\pi;$ (b) $\boldsymbol{c}(t) = \left(\sin 3t, \cos 3t, 2t^{\frac{3}{2}}\right); \quad 0 \le t \le 1.$

57. Parametric Curves. Find a parametrisation for the part of the circle $x^2 + y^2 = a^2$ in the first quadrant of the xy-plane, in terms of the

(a) y-coordinate, oriented anti-clockwise;

(b) x-coordinate, oriented clockwise;

(c) polar angle θ, oriented anti-clockwise;

(d) polar angle θ, oriented clockwise.

58. Path Integrals. Evaluate the following path integrals $\int_C f ds$ where C is given by $\boldsymbol{c}(t)$

(a) $f(x, y, z) = x + y + z, \quad \boldsymbol{c}(t) = (\sin t, \cos t, t), \quad 0 \le t \le 2\pi;$

(b) $f(x, y, z) = e^{\sqrt{z}}, \quad \boldsymbol{c}(t) = (1, 2, t^2), \quad 0 \le t \le 1;$

(c) $f(x, y, z) = \dfrac{x + y}{y + z}, \quad \boldsymbol{c}(t) = \left(t, \dfrac{2}{3}t^{\frac{3}{2}}, t\right), \quad 1 \le t \le 2.$

59. More Path Integrals. (a) Evaluate the path integral

$$\int_C x \cos z \, ds$$

where C is the parabola $y = x^2, z = 0$ from $(0, 0, 0)$ to $(1, 1, 0)$.

(b) Find the mass of a wire along the curve given by

$$\boldsymbol{c}(t) = 3t\boldsymbol{i} + 3t^2\boldsymbol{j} + 2t^3\boldsymbol{k} \quad \text{for} \quad 0 \le t \le 1$$

if the mass density at $\boldsymbol{c}(t)$ is $1 + t$ grams per unit length.

(c) Let the curve C traverse part of the ellipse $9x^2 + y^2 = 9$ from $(0, -3)$ to $(1, 0)$, in a clockwise direction. Determine the total charge of a wire in the shape of C, if the charge density of the wire is $f(x, y) = \sqrt{x^2 + \dfrac{y^2}{81}}$ Coulombs per unit length.

60. Line Integrals of vector fields. Let $\boldsymbol{F}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$. Evaluate the following line integrals $\int_C \boldsymbol{F} \cdot d\boldsymbol{c}$ where

(a) $\boldsymbol{c}(t) = (t, t, t), \quad 0 \leq t \leq 1;$

(b) $\boldsymbol{c}(t) = (\cos t, \sin t, 0), \quad 0 \leq t \leq 2\pi;$

(c) $\boldsymbol{c}(t) = (t^2, 3t, 2t^3), \quad -1 \leq t \leq 2.$

61. More Line Integrals. Evaluate the following integrals.

(a) $\int_C yzdx + xzdy + xydz$ where C consists of straight line segments joining (1,0,0) to (0,1,0) to (0,0,1).

(b) $\int_C x^2dx - xydy + dz$ where C is the parabola $z = x^2, y = 0$ from $(-1, 0, 1)$ to $(1, 0, 1)$.

(c) $\int_C 2xdx + ydy$ where C is the closed curve consisting of the semi circle $y = \sqrt{1 - x^2}$ and the line segment $-1 \leq x \leq 1$ traversed in the anticlockwise direction.

(d) $\int_C ydx - xdy$ where C is the boundary of the square $[-1, 1] \times [-1, 1]$ traversed in the clockwise direction.

62. Line Integral Properties. Let C be a smooth curve in $\mathbb{R}^3$ given by $\boldsymbol{c}(t)$ and $\boldsymbol{F}$ a vector field in $\mathbb{R}^3$.

(a) If $\boldsymbol{F}(\boldsymbol{c}(t))$ is perpendicular to $\boldsymbol{c}'(t)$, show that $\int_C \boldsymbol{F} \cdot d\boldsymbol{c} = 0$.

(b) If $\boldsymbol{F}(\boldsymbol{c}(t))$ is parallel to $\boldsymbol{c}'(t)$ (which means they are in the same direction), show that $\int_C \boldsymbol{F} \cdot d\boldsymbol{c} = \int_C |\boldsymbol{F}| ds$.

MAST20009 Vector Calculus

Problem Sheet 4

Double and Triple Integrals

63. Revision Integrals. Find the indefinite integrals of the following functions.

(a) $\cos^4 x$; (b) $\dfrac{1}{x^2(x+1)}$; (c) $\sin^6 x \cos^3 x$;

(d) $x \cosh x$; (e) $x^2 e^{x^3}$; (f) $\sqrt{1+4x^2}$.

64. Double Integrals Revision. Evaluate the following double integrals.

(a) $\displaystyle\int_0^{\frac{\pi}{2}} \int_0^1 (y \cos x + 2) dy dx$; (b) $\displaystyle\int_{-1}^0 \int_1^2 (-x \log y) dy dx$.

65. Double Integrals. Evaluate the following double integrals.

(a) $\displaystyle\int_0^1 \int_0^{3x} x e^y dy dx$; (b) $\displaystyle\int_0^1 \int_0^y 2x \cos(y^3) dx dy$.

66. Double Integrals. (a) Let R be the region enclosed by the lines $y = 0, y = x$ and $y = 6 - 2x$. Evaluate

$$\iint_R x \, dR.$$

(b) Let D be the interior of the triangle with vertices $(0,0), (1,3)$ and $(2,2)$. Evaluate

$$\iint_D e^{x-y} dA.$$

67. Double Integral Applications. Using double integrals find

(a) the volume of the solid bounded by the graph $z = f(x,y) = 1 + 2x + 3y$, the rectangle $R = [1,2] \times [0,1]$ and the vertical sides of R;

(b) the area of the region enclosed by $x = 1 - y^2$ and $y = -x - 1$;

(c) the area of the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ where $a, b > 0$.

68. Iterated Integrals. For each of the following integrals sketch the region of integration, evaluate the integral as given, and then check your answer by interchanging the order of integration.

(a) $\displaystyle\int_0^2 \int_0^x x^2 y \, dydx$; (b) $\displaystyle\int_0^2 \int_0^{x^2} xy \, dydx$;

(c) $\displaystyle\int_0^4 \int_{\sqrt{y}}^2 (x+y) \, dxdy$.

69. Look Before You Leap. Evaluate the following double integrals.

(a) $\displaystyle\int_0^9 \int_{\sqrt{y}}^3 \sin(x^3) \, dxdy$; (b) $\displaystyle\int_1^e \int_0^{\log x} y \, dydx$.

70. Triple Integrals. Evaluate the following triple integrals.

(a) $\displaystyle\iiint_B ye^{-xy} dxdydz$ where $B = [0,1] \times [0,1] \times [0,1]$;

(b) $\displaystyle\iiint_B (2x+3y+4z) dxdydz$ where $B = [0,2] \times [-1,1] \times [0,1]$;

(c) $\displaystyle\int_0^1 \int_0^x \int_0^y (y+xz) dzdydx$;

(d) $\displaystyle\int_0^1 \int_0^{2x} \int_{x^2+y^2}^{x+y} dzdydx$.

71. Surfaces Revision. Describe and sketch the following surfaces.

(a) $z = \sqrt{1-x^2-y^2}$; (b) $z = 2\sqrt{x^2+y^2}$;

(c) $z = 6-2x-3y$; (d) $(x-1)^2 + y^2 + z^2 = 16$;

(e) $x^2+y^2 = 9$, $-2 \le z \le 4$; (f) $z = 3 - x^2 - y^2$.

72. Elementary Regions. Describe the given region as an elementary region, that is, determine the range of x, y, z specifying the region.

(a) The region between the cone $z = \sqrt{x^2+y^2}$ and the paraboloid $z = x^2+y^2$.

(b) The region inside the sphere $x^2+y^2+z^2 = 1$ and above the plane $z = 0$.

(c) The region cut out of the ball $x^2+y^2+z^2 \le 4$ by the elliptic cylinder $2x^2+y^2 = 1$; that is the region inside the cylinder and the ball.

(d) The region bounded by the elliptic cylinder $x^2+3z^2 = 9$ and the planes $y = 0$ and $y = 5$.

73. Volumes via Triple Integrals. Using triple integrals and cartesian coordinates, find the volume of the

(a) solid bounded by $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ $(a, b, c > 0)$ and the coordinate planes $x = 0, y = 0, z = 0$.

(b) solid bounded by $z = 3x^2 + 3y^2 - 7$ and $z = 9 - x^2 - y^2$.

74. Change of Variables. (a) Let D be the triangle with vertices $(0, 0), (1, 0)$ and $(0, 1)$. Evaluate

$$\iint_D \exp\left(\frac{y - x}{y + x}\right) dxdy$$

by making the substitutions $u = y - x$ and $v = y + x$.

(b) Let D be the region bounded by $x = 0, y = 0, x + y = 1$ and $x + y = 4$. Evaluate

$$\iint_D \frac{dxdy}{x + y}$$

by making the change of variables $x = u - uv, y = uv$.

75. More Change of Variables. Let D be the parallelogram with vertices $(0, 0), (1, 0), (1, 2)$ and $(2, 2)$. Let P be the parallelogram with vertices $(1, 1), (2, 1), (2, 3)$ and $(3, 3)$. Let R be the rectangle $[0, 1] \times [0, 2]$.

(a) Determine a linear transformation that maps D in the x-y plane to R in the u-v plane.

(b) Using the change of variables in part (a), evaluate the double integral

$$\iint_D (x + y) dxdy.$$

(c) By modifying your answer to part (a), determine a transformation that maps P in the xy-plane to R in the uv-plane.

(d) Using the change of variables in part (c), evaluate the double integral

$$\iint_P (x + y) dxdy.$$

76. Polar Coordinates. (a) Let D be the unit disk $x^2 + y^2 \leq 1$. Evaluate

$$\iint_D e^{x^2+y^2}\, dA.$$

(b) Let R be the region bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. Evaluate

$$\iint_R \left(3x + 8y^2\right) dA.$$

77. Elliptical Coordinates. To evaluate double integrals over elliptical regions, it is useful to change variables by modifying the definition of polar coordinates. Let D be the region enclosed by the ellipse $2x^2 + 3y^2 = 1$ and the line $y = 0$, for $y \leq 0$.

Evaluate the double integral

$$\iint_D \sinh\left(4x^2 + 6y^2\right) dxdy$$

by making the change of variables $x = \dfrac{r}{\sqrt{2}} \cos\theta$ and $y = \dfrac{r}{\sqrt{3}} \sin\theta$.

78. Change of Coordinates. Sketch the regions of integration. Write down the integrals obtained after making the required change of coordinates. Do NOT evaluate the integrals.

(a) $\displaystyle\int_{-1}^{1} \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \int_{-\sqrt{4-x^2-y^2}}^{\sqrt{4-x^2-y^2}} xyz\, dzdxdy,$ cylindrical coordinates.

(b) $\displaystyle\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{4-x^2-y^2}} z^2\, dzdxdy,$ spherical coordinates.

79. Change Coordinates. Make an appropriate change of coordinates to evaluate the following integrals.

(a) $\displaystyle\iiint_S \frac{dxdydz}{\left(x^2 + y^2 + z^2\right)^{\frac{3}{2}}}$ where S is the solid bounded by two spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$ where $0 < b < a$.

(b) $\displaystyle\iiint_B z\, dxdydz$ where B is the region within the cylinder $x^2 + y^2 = 1$ that is above the xy plane and below the cone $z = \sqrt{x^2 + y^2}$.

80. Volumes. Using either cylindrical or spherical coordinates, as appropriate, find the volume

(a) inside the surfaces $z = x^2 + y^2$ and $z = \sqrt{2 - x^2 - y^2}$.

(b) of the solid bounded by $z = 3x^2 + 3y^2 - 7$ and $z = 9 - x^2 - y^2$.

81. Centre of Mass. (a) Find the centre of mass of the region between $y = x^2$ and $y = x$ if the mass density is $x + y$.

(b) Find the centre of mass of the solid bounded by $z = 3x^2 + 3y^2 - 7$ and $z = 9 - x^2 - y^2$ if the mass density of the solid is constant.

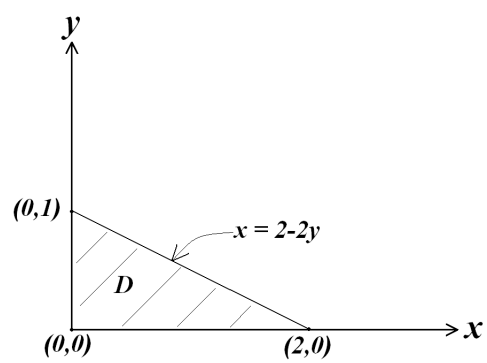
82. Averages. The average value of a quantity f in a domain D in the xy-plane is given by:

$$\overline{f} = \frac{\iint_D f(x,y)\ dxdy}{\iint_D dxdy}$$

The average value of a quantity f in a domain D in xyz-space is given by:

$$\overline{f} = \frac{\iiint_D f(x,y,z)\ dxdydz}{\iiint_D dxdydz}$$

(a) Consider the triangular metal plate in the diagram below. If the thickness of the plate varies as $f(x,y) = (1 + xy)$ cm, find the average thickness of the metal plate.



(b) The temperature at points in the cube $C = [-1,1] \times [-1,1] \times [-1,1]$ is $32d^2$, where d is the distance to the origin

(i) What is the average temperature?
(ii) At what points in the cube is the temperature equal to the average temperature?

83. Moment of Inertia. Find the moment of inertia around the y axis for the ball $x^2 + y^2 + z^2 \leq R^2$ if the mass density is a constant ρ.

84. Another Application. A gold plate D is defined by $0 \leq x \leq 2\pi$ cm and $0 \leq y \leq \pi$ cm and has mass density ρ given by $\rho(x,y) = y^2\sin^2(4x) + 2\,\text{g/cm}^2$. If gold sells for \$7 per gram, how much is the gold plate worth?

MAST20009 Vector Calculus

Problem Sheet 5

Integrals over Surfaces

85. Surface Area. (a) Let $\Phi(u,v) = (u-v, u+v, uv)$ and D be the unit disk in the uv-plane. Find the area of $\Phi(D)$.

(b) Find the surface area of the torus represented parametrically by

$$x = (R + \cos\phi)\cos\theta, \quad y = (R + \cos\phi)\sin\theta, \quad z = \sin\phi$$

where $0 \le \phi \le 2\pi,\ 0 \le \theta \le 2\pi$ and $R > 1$ is fixed.

86. Surface Integrals of Scalar functions. Evaluate the following surface integrals.

(a) $\displaystyle\iint_S z\,dS$ where S is the upper hemisphere ($z \ge 0$) of radius a, centred at $(0,0,0)$.

(b) $\displaystyle\iint_S x\,dS$ where S is the triangular plate with vertices (1,0,0), (0,2,0) and (0,1,1).

(c) $\displaystyle\iint_S z^2\,dS$ where S is the surface of the cube $[-1,1] \times [-1,1] \times [-1,1]$.

87. Mass of Sphere. A metallic surface S is in the shape of a hemisphere

$$z = \sqrt{R^2 - x^2 - y^2}, \quad 0 \le x^2 + y^2 \le R^2.$$

If the mass of the metal per unit area of S is given by $m(x,y,z) = x^2 + y^2$, find the total mass of S.

88. Flux Across a Surface. In (a) - (c) take the normal pointing outwards from the surface.

(a) Find the flux of $\boldsymbol{F}(x,y,z) = 3xy^2\boldsymbol{i} + 3x^2y\boldsymbol{j} + z^3\boldsymbol{k}$ out of the unit sphere centred at $(0,0,0)$.

(b) A uniform fluid that flows vertically downward is described by the vector field $\boldsymbol{F}(x,y,z) = -\boldsymbol{k}$. Find the flux through the cone $z = \sqrt{x^2+y^2}, \quad x^2 + y^2 \le 1$.

(c) Let S be the closed surface that consists of the upper hemisphere ($z \ge 0$) of radius 1, centred at $(0,0,0)$, and its base $x^2 + y^2 \le 1, \quad z = 0$. Let the electric field be defined by $\boldsymbol{E}(x,y,z) = 2x\boldsymbol{i} + 2y\boldsymbol{j} + 2z\boldsymbol{k}$. Find the electric flux across S.

89. Surface Integrals of Vector fields. Evaluate the following surface integrals.

(a) $\iint_S \boldsymbol{F} \cdot d\boldsymbol{S}$ where $\boldsymbol{F}(x, y, z) = 18z\boldsymbol{i} - 12\boldsymbol{j} + 3y\boldsymbol{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ where $x \geq 0, y \geq 0, z \geq 0$. Take the normal pointing upwards.

(b) $\iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot d\boldsymbol{S}$ where $\boldsymbol{F}(x, y, z) = (x^2 + y - 4)\boldsymbol{i} + 3xy\boldsymbol{j} + (2xz + z^2)\boldsymbol{k}$ and S is the upper hemisphere ($z \geq 0$) of radius 4, centred at $(0, 0, 0)$. Take the normal pointing upwards.

MAST20009 Vector Calculus

Problem Sheet 6

Integral Theorems

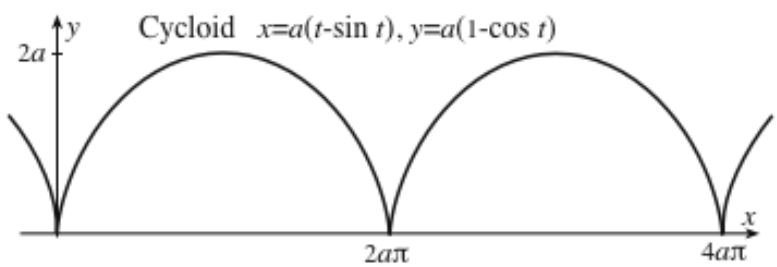
90. Green's Theorem. Verify Green's theorem for the disk D with centre (0,0) and radius R and the functions P and Q.

(a) $P(x,y) = xy^2, \quad Q(x,y) = -yx^2;$ (b) $P(x,y) = 2y, \quad Q(x,y) = x.$

91. Apply Green's Theorem. Redo questions 61(c) and 61(d) using Green's theorem.

92. Area via Line Integrals. Find the area of the following regions using Green's theorem and an appropriate line integral.

(a) The region bounded by one arc of the cycloid $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$ where $a > 0$ and $0 \le \theta \le 2\pi$ and the x axis.



(b) The region inside the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ where $a, b > 0$.

93. Harmonic Functions. Let D be a region for which Green's theorem holds and ∂D is traversed anticlockwise. Suppose that f is harmonic, that is $f_{xx} + f_{yy} = 0$ on D. Prove that

$$\int_{\partial D} \frac{\partial f}{\partial y} dx - \frac{\partial f}{\partial x} dy = 0.$$

94. Divergence Theorem in Plane. Verify the divergence theorem in the plane for the

(a) unit disk $x^2 + y^2 \le 1$ if $\boldsymbol{F}(x,y) = x\boldsymbol{i} + y\boldsymbol{j}$;

(b) unit square $[0,1] \times [0,1]$ if $\boldsymbol{F}(x,y) = 10x\boldsymbol{i} - 5y\boldsymbol{j}$.

95. Apply Divergence Theorem in Plane. Let $\hat{\boldsymbol{n}}$ be the unit outward normal to the curve C in the xy-plane. Evaluate

$$\int_C \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\, ds$$

where

(a) $\boldsymbol{F}(x,y) = \big(2\sin y + e^y + 3x, 3\cos x - e^{4x} - y\big)$ and C is the rectangle $[-1,2]\times[0,4]$ traversed anticlockwise.

(b) $\boldsymbol{F}(x,y) = \left(x^2 + 5y + y^3, \frac{1}{2}y^2 + 3x - x^5\right)$ and C is the triangle with vertices $(-1,0), (0,0), (0,2)$ traversed anticlockwise.

96. Integral Theorem Proof. Let D be a region in the xy-plane bounded by a simple closed curve C which is traversed anticlockwise. Let $\phi(x,y)$ and $\psi(x,y)$ be C^2 scalar functions. Let $\hat{\mathbf{n}}$ be the unit outward normal to C in the xy-plane. Prove that

$$\int_C (\phi\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\phi)\cdot\hat{\mathbf{n}}\, ds = \iint_D \big(\phi\nabla^2\psi - \psi\nabla^2\phi\big)\, dxdy.$$

97. Stokes' Theorem. Verify Stokes' theorem for the upper hemisphere of radius 1, centred at $(0,0,0)$, where the vector field is

(a) $\boldsymbol{F}(x,y,z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$; (b) $\boldsymbol{F}(x,y,z) = \left(x^3 + \frac{z^4}{4}, 4x, xz^3 + z^2\right)$.

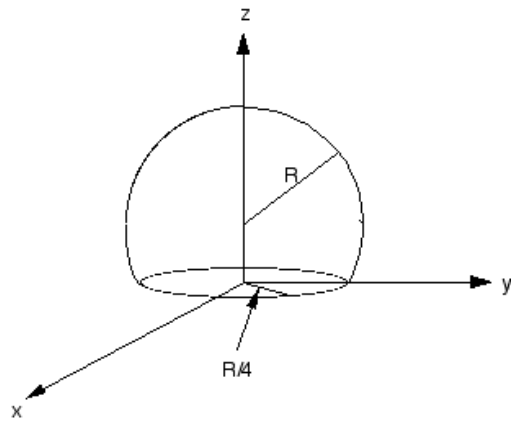
98. Apply Stokes' Theorem. (a) Redo question 89(b) using Stokes' theorem.

(b) Using Stokes' theorem evaluate

$$\iint_S (\boldsymbol{\nabla}\times\boldsymbol{F})\cdot d\boldsymbol{S}$$

where $\boldsymbol{F}(x,y,z) = y\boldsymbol{i} - x\boldsymbol{j} + zx^3y^2\boldsymbol{k}$ and S is the surface $x^2+y^2+3z^2 = 1, z \le 0$. Take normal pointing downwards.

(c) A hot air balloon has the truncated spherical shape shown below. The hot gases escape through the porous surface with a velocity field $\boldsymbol{F}(x,y,z) = \boldsymbol{\nabla}\times(-y,x,0)$. If $R = 5$, compute the volume flow rate (flux) of the gases through the surface. Take normal pointing outwards from balloon.



99. Integral Proofs. If C is a closed curve which is the boundary of an open surface S, and f and g are C^2 functions, show that

$$\text{(a)} \int_C f\boldsymbol{\nabla} g \cdot d\mathbf{c} = \iint_S (\boldsymbol{\nabla} f \times \boldsymbol{\nabla} g) \cdot d\boldsymbol{S}; \qquad \text{(b)} \int_C (f\boldsymbol{\nabla} g + g\boldsymbol{\nabla} f) \cdot d\boldsymbol{c} = 0.$$

100. Conservative Fields. In each case, show that $\boldsymbol{F}$ is a conservative vector field and find a scalar function ϕ such that $\boldsymbol{F} = \boldsymbol{\nabla}\phi$. Evaluate $\int_C \boldsymbol{F} \cdot d\boldsymbol{c}$ along paths joining $(1, -2, 1)$ to $(3, 1, 4)$.

$$\text{(a) } \boldsymbol{F}(x, y, z) = (2xyz + \sin x)\boldsymbol{i} + x^2 z\boldsymbol{j} + x^2 y\boldsymbol{k};$$

$$\text{(b) } \boldsymbol{F}(x, y, z) = (2xy + z^3)\boldsymbol{i} + x^2\boldsymbol{j} + 3xz^2\boldsymbol{k}.$$

101. Further Line Integrals. Evaluate $\int_C \boldsymbol{F} \cdot d\boldsymbol{c}$ along the curve C given by $\boldsymbol{c}(t)$.

(a) $\boldsymbol{F}(x, y, z) = e^x \sin y\boldsymbol{i} + e^x \cos y\boldsymbol{j} + z^2\boldsymbol{k}$; $\quad \boldsymbol{c}(t) = (\sqrt{t}, t^3, e^{\sqrt{t}}), \quad 0 \le t \le 1.$

(b) $\boldsymbol{F}(x, y, z) = (xy^2 + 3x^2 y)\boldsymbol{i} + (x^3 + x^2 y)\boldsymbol{j}$; $\quad C$ is the path consisting of line segments from (1,1) to (0,2) to (3,0).

102. Divergence Theorem. Verify Gauss' Divergence theorem for the cube $[0, 1] \times [0, 1] \times [0, 1]$ when

$$\boldsymbol{F}(x, y, z) = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}.$$

103. Apply Divergence Theorem. (a) Redo questions 88(a) and 88(c) using Gauss' Divergence theorem.

(b) Let S be the surface of the closed region $x^2 + y^2 \le z \le 1$ and $x \ge 0$. Find the flux of the velocity field $\boldsymbol{F}(x, y, z) = (x - y, y - z, z - x)$ in the direction of the outward normal to S.

104. More Integral Proofs. Let $\boldsymbol{F}$ be a smooth vector field in $\mathbb{R}^3$. Let ∂W be an oriented closed surface that bounds a region W.

(a) Suppose $\boldsymbol{F}$ is tangent to ∂W. Prove that

$$\iiint_W \boldsymbol{\nabla} \cdot \boldsymbol{F}\ dV = 0.$$

(b) Prove that

$$\iiint_W (\boldsymbol{\nabla} f) \cdot \boldsymbol{F}\ dV = \iint_{\partial W} f\boldsymbol{F} \cdot d\boldsymbol{S} - \iiint_W f\boldsymbol{\nabla} \cdot \boldsymbol{F}\ dV.$$

105. Mixed Integral Theorems. In (a), (b), (d), and (e) take normal pointing outwards from the surface. By applying an appropriate integral theorem, evaluate the following integrals.

(a) $\displaystyle\iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot d\boldsymbol{S}$ over the portion of the surface $2z = x^2 + y^2$ below the plane $z = 2$ when $\boldsymbol{F}(x, y, z) = (3y, -xz, -yz^2)$.

(b) $\displaystyle\iint_S \boldsymbol{F} \cdot d\boldsymbol{S}$ where S is the surface of the closed cylinder $x^2 + y^2 \leq 1, \quad 0 \leq z \leq 1$ and $\boldsymbol{F}(x, y, z) = (1, 1, z(x^2 + y^2)^2)$.

(c) $\displaystyle\int_C x^3 dy - y^3 dx$ where C is the unit circle $x^2 + y^2 = 1$ traversed in the clockwise direction.

(d) $\displaystyle\iint_S \boldsymbol{F} \cdot d\mathbf{S}$ where $\boldsymbol{F}(x, y, z) = (2xy + z, y^2, -x - 3y)$ and S is surface of the tetrahedron bounded by $2x + 2y + z = 6, x = 0, y = 0, z = 0$.

(e) $\displaystyle\iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot d\boldsymbol{S}$ where $\boldsymbol{F}(x, y, z) = (zx + z^2y + x, z^3yx + y, z^4x^2)$. Let S be the capped cylindrical surface given by the union of two surfaces S_1 and S_2 where S_1 is $x^2+y^2 = 1, 0 \leq z \leq 1$ and

S_2 is $x^2 + y^2 + (z - 1)^2 = 1, z \geq 1$.

(f) $\displaystyle\int_C \boldsymbol{F} \cdot \hat{\boldsymbol{n}}\ ds$ where $\boldsymbol{F}(x, y) = 2xy\boldsymbol{i} - y^2\boldsymbol{j}$, $\hat{\boldsymbol{n}}$ is the unit outward normal to the curve C in the xy-plane and C is the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ $(a, b > 0)$, traversed in the anticlockwise direction.

106. Think Carefully. In (a) and (b) take the normal pointing outwards from the surface. In each case, use an appropriate integral theorem to evaluate the integral.

(a) $\iint_S \boldsymbol{F} \cdot d\boldsymbol{S}$ where $\boldsymbol{F}(x, y, z) = (x^3, 3yz^2, 3y^2z + 10)$ and S is the surface $z = -\sqrt{4 - x^2 - y^2}$.

(b) $\iint_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot d\boldsymbol{S}$ where S is that part of the sphere $x^2 + y^2 + (z - 2)^2 = 5$ for $z \geq 1$ and $\boldsymbol{F}(x, y, z) = (e^{y+z} - y^3\,, xe^{y+z}\,, e^{-xyz} - z^2\,)\,.$

(c) $\int_C \boldsymbol{F} \cdot d\boldsymbol{c}$ where $\boldsymbol{F}(x, y) = \dfrac{y}{x^2 + y^2}\boldsymbol{i} - \dfrac{x}{x^2 + y^2}\boldsymbol{j}$ and C is the curve $x^2 + 2y^2 = 1$ oriented anticlockwise.

107. Maxwells Equations. Verify that Maxwell's equations imply the equation of continuity for $\boldsymbol{J}$ and ρ.

Hint: Take the divergence of Ampere's Law and differentiate Gauss' Law.

MAST20009 Vector Calculus

Problem Sheet 7

General Curvilinear Coordinates

108. Elliptical Cylindrical Coordinates. Define curvilinear coordinates (u, v, z) by

$$x = a\cosh u \cos v, \qquad y = a\sinh u \sin v, \qquad z = z$$

where $u \geq 0, 0 \leq v \leq 2\pi$ and $a > 0$ is a real constant.

(a) Find the unit vectors $\boldsymbol{e}_u, \boldsymbol{e}_v, \boldsymbol{e}_z$.

(b) Show that the system is orthogonal.

(c) Show that the element of volume is of the form

$$dV = a^2(\sinh^2 u + \sin^2 v)dudvdz.$$

109. Cartesian to Cylindrical. Represent the vector $\mathbf{A}(x, y, z) = z\boldsymbol{i} - 2x\boldsymbol{j} + y\boldsymbol{k}$ in cylindrical coordinates by writing it in the form

$$\mathbf{A}(\rho, \phi, z) = A_1(\rho, \phi, z)\hat{\boldsymbol{\rho}} + A_2(\rho, \phi, z)\hat{\boldsymbol{\phi}} + A_3(\rho, \phi, z)\hat{\boldsymbol{z}}.$$

110. Cylindrical Coordinates. Define curvilinear coordinates (ρ, ϕ, z) by

$$x = \rho\cos\phi, \qquad y = \rho\sin\phi, \qquad z = z.$$

where $\rho \geq 0, 0 \leq \phi < 2\pi$. If n is an integer, find expressions for the following quantities in terms of ρ, ϕ, z and $\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}, \hat{\boldsymbol{z}}$.

(a) $\boldsymbol{\nabla}\phi$; (b) $\boldsymbol{\nabla}\rho^n$;

(c) $\nabla^2(\rho^2\cos\phi)$; (d) $\boldsymbol{\nabla}\cdot(\rho\hat{\boldsymbol{\rho}} + \rho\phi\hat{\boldsymbol{\phi}} + z\hat{\boldsymbol{z}})$;

(e) $\boldsymbol{\nabla}\times\hat{\boldsymbol{\phi}}$; (f) $\boldsymbol{\nabla}\cdot(\rho^{n-1}\sin(n\phi)\hat{\boldsymbol{\rho}} + \rho^{n-1}\cos(n\phi)\hat{\boldsymbol{\phi}})$.

111. Spherical Coordinates. Define curvilinear coordinates (r, θ, ϕ) by

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta.$$

where $r \geq 0, 0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi$. Find expressions for the following quantities in terms of r, θ, ϕ and $\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$.

(a) $\boldsymbol{\nabla}\phi$; (b) $\boldsymbol{\nabla}\theta$;

(c) $\boldsymbol{\nabla}\cdot(\hat{\boldsymbol{r}}\cot\phi - 2\hat{\boldsymbol{\phi}})$; (d) $\boldsymbol{\nabla}\times(\hat{\boldsymbol{r}}\cot\phi - 2\hat{\boldsymbol{\phi}})$;

(e) $\nabla^2(r + \theta + \phi)$; (f) $\boldsymbol{\nabla}\cdot(\sin\theta\hat{\boldsymbol{\theta}})$.

MAST20009 Vector Calculus
Problem Sheet 1
Functions of Several Variables

1. Limits.

(a) 4 (b) 0 (c) does not exist (d) 5

(e) does not exist (f) does not exist (g) 0

(h) does not exist (i) 1 (j) does not exist

2. No Limits.

3. Limit Exists.

4. Continuity.

(a) (i) is continous at $(x, y) = (0, 0)$.

(b) (ii) can be made continuous by defining $f(x, y) = 1$ at $(x, y) = (0, 0)$.

5. Partial Derivatives Revision.

In order listed.

(a) $\dfrac{1}{x+3y}, \quad \dfrac{3}{x+3y}, \quad \dfrac{-1}{(x+3y)^2}, \quad \dfrac{-3}{(x+3y)^2}, \quad \dfrac{-3}{(x+3y)^2}, \quad \dfrac{-9}{(x+3y)^2}$

(b) $ye^{xy}, \quad xe^{xy}, \quad y^2e^{xy}, \quad e^{xy} + xye^{xy}, \quad e^{xy} + xye^{xy}, \quad x^2e^{xy}$

(c) $2xy^3z^5, \quad 3x^2y^2z^5, \quad 5x^2y^3z^4, \quad 6x^2yz^5, \quad 15x^2y^2z^4, \quad 30xy^2z^4$

(d) $\dfrac{1}{y}, \quad \dfrac{-x}{y^2} + \dfrac{1}{z}, \quad \dfrac{-y}{z^2}, \quad \dfrac{-1}{z^2}, \quad \dfrac{2x}{y^3}, \quad \dfrac{2}{y^3}$

6. Laplace's Equation.

$$V_{xx} = GmM\frac{y^2 + z^2 - 2x^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}, \quad V_{yy} = GmM\frac{x^2 + z^2 - 2y^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}, \quad V_{zz} = GmM\frac{x^2 + y^2 - 2z^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}$$

7. Differentiable Functions.

(a) $x \neq 0, y \neq 0$ (b) $(x, y) \neq (0, 0)$ (c) $r > 0$, all θ

8. Continuity and Differentiability.

(a) $\dfrac{\partial f}{\partial y} = \dfrac{-3x^2y^2 - 5y^4 - 40x^2y}{(x^2+5y^2)^2}$ (b) $-\dfrac{1}{5}$

(c) No, since $\lim_{(x,y)\to(0,0)} \frac{\partial f}{\partial y}$ does not exist (d) No

9. Different Mixed Partials.

(a) $\dfrac{\partial f}{\partial x} = \dfrac{x^4y - y^5 + 4x^2y^3}{(x^2+y^2)^2}, \quad \dfrac{\partial f}{\partial y} = \dfrac{x^5 - 4x^3y^2 - xy^4}{(x^2+y^2)^2}$

(c) $f(x,y)$ is not C^2

10. Matrix Chain Rule.

(a) 24 (b) $(8,-16)$ (c) $\begin{bmatrix} -3 & 2 \\ 12 & 24 \end{bmatrix}$

11. Matrix Chain Rule Again.

(a) $\begin{bmatrix} -2 & 2 \\ 0 & 16 \\ 1 & -9 \end{bmatrix}$ (b) $\begin{bmatrix} -256 & 0 \\ 512 & 0 \end{bmatrix}$

12. Cylindrical Coordinates.

(a) ρ (b) $\rho = \sqrt{X^2+Y^2}, \quad \tan\phi = \dfrac{Y}{X}$

(c) $\dfrac{\partial \rho}{\partial X} = \cos\phi, \quad \dfrac{\partial \phi}{\partial Y} = \dfrac{\cos\phi}{\rho}, \quad \dfrac{\partial z}{\partial Z} = 1$

13. Spherical Coordinates.

(a) $r^2\sin\theta$ (b) $r = \sqrt{x^2+y^2+z^2}, \theta = \arccos\left(\dfrac{z}{r}\right), \quad \tan\phi = \dfrac{y}{x}$

(c) $\dfrac{\partial r}{\partial x} = \sin\theta\cos\phi, \quad \dfrac{\partial \theta}{\partial y} = \dfrac{\sin\phi\cos\theta}{r}, \quad \dfrac{\partial \phi}{\partial z} = 0$

14. Product of Jacobians.

15. Taylor Polynomials.

(a) $1 + x + y + \dfrac{1}{2}(x^2+2xy+y^2) + \dfrac{1}{6}\left(x^3+3x^2y+3xy^2+y^3\right)$

(b) $1 + xy$

(c) $1 - (x-1) + (y-1) + (x-1)^2 - 2(x-1)(y-1) + (y-1)^2$

(d) $-4\left(x-\frac{\pi}{8}\right)+3y+\frac{1}{6}\left[64\left(x-\frac{\pi}{8}\right)^3-144\left(x-\frac{\pi}{8}\right)^2 y+108\left(x-\frac{\pi}{8}\right)y^2-27y^3\right]$

16. Taylor Polynomial Application.

(a) $1+(x-1)^2-\frac{1}{2}y^2$
(b) approximate answer is 0.99, exact answer is 0.98992
(c) $\frac{7}{6}$

17. Taylor Remainder.

(a) $1+\frac{1}{4}x+\frac{1}{2}y$ (b) 1.075, $|R_1|\leq 0.0015$

18. Another Taylor Remainder.

(a) $1-2(x-1)+(y+1)+2(x-1)^2-2(x-1)(y+1)+\frac{1}{2}(y+1)^2$, 1.105
(b) $\frac{e^{0.1}}{6000}<0.000184$

19. Maxima and Minima Revision.

(a) saddle point at $(0,0)$
(b) minimum at $(2,1)$, saddle point at $(0,0)$
(c) mimimum on $y=-x$.

20. Eliminate Variable.

(a) $F(x,y)=x^2-y^2+y^3$
(b) minimum at $\left(0,\frac{2}{3}\right)$, saddle point at $(0,0)$.
(c) minimum at $\left(0,\frac{2}{3},\frac{4}{9}\right)$, saddle point at $(0,0,0)$.

21. Use Lagrange Multipliers.

$\left(0,\frac{2}{3},\frac{4}{9}\right)$, $(0,0,0)$.

22. Lagrange Multipliers.

(a) maximum at $\left(\sqrt{\frac{2}{3}},-\sqrt{\frac{2}{3}},\sqrt{\frac{2}{3}}\right)$, minimum at $\left(-\sqrt{\frac{2}{3}},\sqrt{\frac{2}{3}},-\sqrt{\frac{2}{3}}\right)$
(b) maximum at $(\sqrt{3},0)$, minimum at $(-\sqrt{3},0)$,
(c) no extrema
(d) maximum at $(\frac{-1}{\sqrt{6}},\frac{2}{\sqrt{6}},1+\frac{2}{\sqrt{6}})$, minimum at $(\frac{1}{\sqrt{6}},\frac{-2}{\sqrt{6}},1-\frac{2}{\sqrt{6}})$.
(e) maximum at $(2,-1,1)$, minimum at $(-2,1,-1)$.

23. More Lagrange Multipliers.

(a) Problem is to minimise $S = 2\pi r^2 + 2\pi rh$ given $\pi r^2 h = 1000$.

radius $= \dfrac{10}{(2\pi)^{\frac{1}{3}}}$ cm, height $= \dfrac{20}{(2\pi)^{\frac{1}{3}}}$ cm

(b) Problem is to minimise $F = x^2 + y^2 + z^2$ given $x + y = 1$ and $x^2 + 2y^2 + z^2 = 1$.

$(x, y, z) = \left(\dfrac{1}{3}, \dfrac{2}{3}, 0\right)$

24. Inequality Constraints.

(a) $f = 8$ is minimum at $(0, 1), (0, -1)$, $f = 15$ is maximum at $(-1, 0), (1, 0)$

(b) Problem is to maximise $V = lwh$ given $l + 2w + 2h \leq 108$.

volume $= 11664$ cm^3

MAST20009 Vector Calculus

Problem Sheet 2

Space Curves and Vector Fields

25. Vectors Revision.

(a) (i) -1 (ii) $(1, -2, -2)$ (b) $\frac{1}{\sqrt{78}}(7, -2, -5)$ (c) $b = 3$.

26. Equation of Lines Revision.

(a) $\boldsymbol{r} = (0, 1, 2) + t(1, -2, 1)$; $x = t,\ y = 1 - 2t,\ z = 2 + t$
(b) $\boldsymbol{r} = (1, 2, 1) + t(3, -4, 2)$; $x = 1 + 3t,\ y = 2 - 4t,\ z = 1 + 2t$

27. Equation of Planes Revision.

(a) $2x - 3y - 7z = -8$; (b) $x + 2y - z = -1$.

28. Parametric curves.
(a) $v = 2\boldsymbol{j},\ a = -\boldsymbol{i}$, tangent$= \boldsymbol{i} + 2t\boldsymbol{j}$
(b) $v = \sqrt{2}\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k},\ a = \boldsymbol{j} + \boldsymbol{k}$, tangent$= \sqrt{2}t\boldsymbol{i} + (1 + t)\boldsymbol{j} + (1 - t)\boldsymbol{k}$

29. Rolling Wheel.

$t = \frac{nR\pi}{v}$ where n is an integer. If n is even speed is 0. If n is odd speed is $2v$.

30. Velocity Properties.

31. Differentiation.

(a) $\mathbf{F} \cdot \frac{d\mathbf{F}}{dt} \times \frac{d^3\mathbf{F}}{dt^3}$ (b) $\mathbf{u}''' \cdot \mathbf{u} \times \mathbf{u}'$

32. Curve Properties.

(a) $\mathbf{T} = (\sin t, \cos t, 0)$, $\mathbf{N} = (\cos t, -\sin t, 0)$, $\mathbf{B} = (0, 0, -1)$,
(b) $\kappa = \frac{1}{t}, \tau = 0$

33. More curve Properties.

(a) $\mathbf{T} = \frac{1}{\sqrt{5}}(2\sin t + \cos t, 2\cos t - \sin t, 0)$,
$\mathbf{N} = \frac{1}{\sqrt{5}}(2\cos t - \sin t, -2\sin t - \cos t, 0)$, $\quad \mathbf{B} = (0, 0, -1)$,
(b) $\kappa = \frac{e^{-2t}}{\sqrt{5}}, \tau = 0$

34. Curve Properties.

35. Parametrised Normals.

(a) (x, y, z), unit sphere centred at origin
(b) $(-x, 0, -z)$ cylinder with axis on y axis extending from $-1 \leq y \leq 3$

36. Parametric Surface 1.

(a) $(4uv, -4v, 2)$ (b) smooth for all u, v (c) $2y - z = 1$

37. Parametric Surface 2.

(a) $(4 - 2u, -8u - 4uv, 2u + 2v)$ (b) not smooth at $(u, v) = (2, -2)$ (c) $4x + z = 1$

38. Convert from Polar.

(a)(i) $(x, y) = \left(\dfrac{-7}{2}, \dfrac{7\sqrt{3}}{2}\right)$ (ii) $(x, y) = \left(-3\sqrt{3}, -3\right)$ (iii) $(x, y) = (0, 0)$
(b)(i) $(r, \theta) = \left(4, \dfrac{11\pi}{6}\right)$ (ii) $(r, \theta) = \left(2, \dfrac{3\pi}{2}\right)$ (iii) $(r, \theta) = \left(8\sqrt{2}, \dfrac{5\pi}{4}\right)$

39. Convert from Cylindrical.

(a) $(x, y, z) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1\right)$, $\quad (r, \theta, \phi) = \left(\sqrt{2}, \frac{\pi}{4}, \frac{\pi}{4}\right)$
(b) $(x, y, z) = (0, 2, -4)$, $\quad (r, \theta, \phi) = \left(2\sqrt{5}, 2.678, \frac{\pi}{2}\right)$
(c) $(x, y, z) = \left(\frac{3\sqrt{3}}{2}, \frac{3}{2}, 4\right)$, $\quad (r, \theta, \phi) = \left(5, 0.644, \frac{\pi}{6}\right)$
(d) $(x, y, z) = \left(-\sqrt{2}, \sqrt{2}, -2\right)$, $\quad (r, \theta, \phi) = \left(2\sqrt{2}, \frac{3\pi}{4}, \frac{3\pi}{4}\right)$

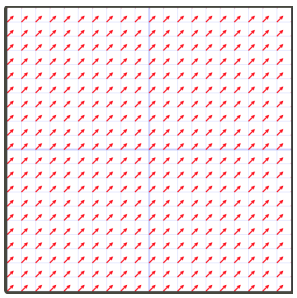
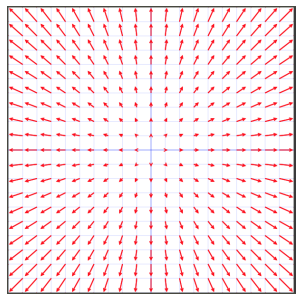
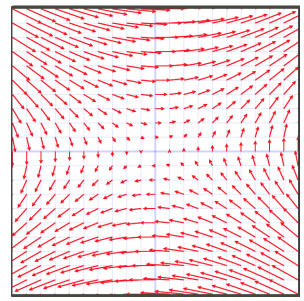
40. Convert from Cartesian.

(a) $(r, \theta, \phi) = (3, 2.301, 0.464)$, $\quad (\rho, \phi, z) = \left(\sqrt{5}, 0.464, -2\right)$
(b) $(r, \theta, \phi) = \left(5, 0.644, \frac{\pi}{2}\right)$, $\quad (\rho, \phi, z) = \left(3, \frac{\pi}{2}, 4\right)$
(c) $(r, \theta, \phi) = \left(2, \frac{\pi}{3}, 0.615\right)$, $\quad (\rho, \phi, z) = \left(\sqrt{3}, 0.615, 1\right)$
(d) $(r, \theta, \phi) = \left(5, 0.927, \frac{7\pi}{6}\right)$, $\quad (\rho, \phi, z) = \left(4, \frac{7\pi}{6}, 3\right)$

41. Parametric Surfaces.

(a) $x = \sqrt{2}\cos\phi, y = y, z = \sqrt{2}\sin\phi,\ 0 \le \phi \le 2\pi,\ 0 \le y \le 2$
(b) $x = \rho\cos\phi,\ y = \rho\sin\phi,\ z = 15 - 3\rho,\ 0 \le \phi \le 2\pi,\ 0 \le \rho \le 5$
(c) $x = 3\sin\theta\cos\phi,\ y = 3\sin\theta\sin\phi,\ z = 3\cos\theta,\ 0 \le \phi \le 2\pi,\ \frac{\pi}{2} \le \theta \le \pi$

42. Vector Fields.

(a)  (b)  (c) 

43. Flow Lines.

(a) $x^2 + y^2 = r^2$ where r is a constant
(b) $y = \frac{c}{x}$ where c is a constant
(c) level curves are concentric circles $x^2 + y^2 = r^2$ where r is a constant

44. Flow Line Properties.

45. Divergence.

(a) 0 (b) 3 (c) $y\cos(xy) + x^2\sin(x^2y)$

46. Curl.

(a) $\cos z\boldsymbol{i} - x\boldsymbol{k}$ (b) $\mathbf{0}$ (c) $(10y - 8z)\boldsymbol{i} - (10x - 6z)\boldsymbol{j} + (8x - 6y)\boldsymbol{k}$

47. Div, Grad, and Curl.

(a) $2xy\boldsymbol{i} + x^2\boldsymbol{j}$ (b) $3xy^2z\boldsymbol{i} + (4xz - y^3z)\boldsymbol{j}$ (c) $2z^2 + xy^3$
(d) undefined (e) $-y^3zx^3\boldsymbol{i} + 2y^4x^2z\boldsymbol{j} + (2x^3z^2 - 2xy)\boldsymbol{k}$ (f) $4x^2yz^2 + x^2$
(g) undefined (h) 2y (i) $4x\boldsymbol{i} + 6xyz\boldsymbol{k}$

48. Curl of Grad.

49. Scalar Potentials.

(a) yes, $\phi = x^2 + \frac{3}{2}y^2 + 2z^2 + c$ (b) no (c) yes, $\phi = x^2ye^z + \frac{1}{3}z^3 + c$

50. Finding Vector Fields.

(a) yes, $\mathbf{G} = -\frac{1}{2}(yz^2 + y^2)\mathbf{i} - \frac{1}{2}xz^2\mathbf{j}$
(b) yes, $\mathbf{G} = -(z\sin y + y\sin x)\mathbf{i} - xz\cos y\mathbf{j}$ (c) no

51. Vector Identity One.

(b) $\frac{\mathbf{r}e^{r^2}}{r^3}(2r^2 - 1)$

52. Vector Identity Two.

(b) $\dfrac{-2}{r^5}$

53. Vector Identity Three.

(b) $\mathbf{0}$

54. Using Vector Identities.

55. Vector Properties.

MAST20009 Vector Calculus

Problem Sheet 3

Integrals over Paths

56. Arc Length.

(a) $2\pi\sqrt{5}$ (b) $2(2\sqrt{2}-1)$

57. Parametric Curves.

(a) $\mathbf{c} = \sqrt{a^2-y^2}\mathbf{i} + y\mathbf{j}, \quad 0 \le y \le a;$ (b) $\mathbf{c} = x\mathbf{i} + \sqrt{a^2-x^2}\mathbf{j}, \quad 0 \le x \le a;$
(c) $\mathbf{c} = a\cos\theta\mathbf{i} + a\sin\theta\mathbf{j}, \quad 0 \le \theta \le \dfrac{\pi}{2};$ (d) $\mathbf{c} = a\cos\theta\mathbf{i} - a\sin\theta\mathbf{j}, \quad \dfrac{3\pi}{2} \le \theta \le 2\pi.$

58. Path Integrals.

(a) $2\sqrt{2}\pi^2$ (b) 2 (c) $\frac{2}{3}(8-3\sqrt{3})$

59. More Path Integrals.

(a) $\frac{1}{12}(5\sqrt{5}-1)$ (b) 8 grams (c) $\frac{5\pi}{2}$ Coulombs

60. Line Integrals of vector fields.

(a) $\frac{3}{2}$ (b) 0 (c) 147

61. More Line Integrals.

(a) 0 (b) $\frac{2}{3}$ (c) 0 (d) 8

62. Line Integral Properties.

MAST20009 Vector Calculus

Problem Sheet 4

Double and Triple Integrals

63. Revision Integrals.

(a) $\frac{1}{32}\sin 4x + \frac{3}{8}x + \frac{1}{4}\sin 2x + C$

(b) $-\log|x| - \frac{1}{x} + \log|x+1| + C$

(c) $\frac{1}{7}\sin^7 x - \frac{1}{9}\sin^9 x + C$

(d) $x\sinh x - \cosh x + C$

(e) $\frac{1}{3}e^{x^3} + C$

(f) $\frac{1}{4}(\operatorname{arcsinh} 2x + 2x\sqrt{1+4x^2}) + C$

64. Double Integrals Revision.

(a) $\frac{1}{2} + \pi$ (b) $\log 2 - \frac{1}{2}$

65. Double Integrals.

(a) $\frac{2}{9}e^3 - \frac{7}{18}$ (b) $\frac{1}{3}\sin 1$

66. Double Integrals.

(a) 5 (b) $1 + e^{-2}$

67. Double Integral Applications.

(a) $\frac{11}{2}$ (b) 4.5 (c) πab

68. Iterated Integrals.

(a) $\dfrac{16}{5}$ (b) $\dfrac{16}{3}$ (c) $\dfrac{36}{5}$

69. Look Before You Leap.

(a) $\frac{1}{3}(1 - \cos 27)$ (b) $\frac{e}{2} - 1$

70. Triple Integrals.

(a) e^{-1} (b) 16 (c) $\frac{7}{60}$ (d) $\frac{1}{6}$

71. Surfaces Revision.

(a) hemisphere radius 1 centred at the origin for $z \geq 0$
(b) cone with the $z-$axis as the axis of symmetry
(c) plane intercepts $(3,0,0),(0,2,0),(0,0,6)$
(d) sphere radius 4 centred at $(1,0,0)$
(e) cylinder of radius 3 and height 6 units with the $z-$axis as the axis of symmetry
(f) inverted parabolic bowl, z intercept (0, 0, 3), with the $z-$axis as the axis of symmetry

72. Elementary Regions.

(a) $-1 \leq x \leq 1, \quad -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}, \quad x^2+y^2 \leq z \leq \sqrt{x^2+y^2}$
(b) $-1 \leq x \leq 1, \quad -\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}, \quad 0 \leq z \leq \sqrt{1-x^2-y^2}$
(c) $\frac{-1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}, \quad -\sqrt{1-2x^2} \leq y \leq \sqrt{1-2x^2}, \quad -\sqrt{4-x^2-y^2} \leq z \leq \sqrt{4-x^2-y^2}$
(d) $-\sqrt{9-3z^2} \leq x \leq \sqrt{9-3z^2}, \quad 0 \leq y \leq 5, \quad -\sqrt{3} \leq z \leq \sqrt{3}$

73. Volumes via Triple Integrals.

(a) $\dfrac{abc}{6}$ (b) 32π

74. Change of Variables.

(a) $\dfrac{1}{4}\left(e-\dfrac{1}{e}\right)$ (b) 3

75. More Change of Variables.

(a) $x=u+\frac{1}{2}v, y=v$ (b) 4 (c) $x=u+\frac{1}{2}v+1, y=v+1$ (d) 8

76. Polar Coordinates.

(a) $\pi(e-1)$ (b) 30π

77. Elliptical Coordinates.

$\frac{\pi}{4\sqrt{6}}(\cosh(2)-1)$

78. Change of Coordinates.

(a) $\displaystyle\int_0^1 \int_0^{2\pi} \int_{-\sqrt{4-\rho^2}}^{\sqrt{4-\rho^2}} z\rho^3 \cos\phi \sin\phi \, dz d\phi d\rho$ (b) $\displaystyle\int_0^2 \int_0^{2\pi} \int_0^{\frac{\pi}{4}} r^4 \cos^2\theta \sin\theta \, d\theta d\phi dr$

79. Change Coordinates.

(a) $4\pi \log\left(\frac{a}{b}\right)$ (b) $\frac{\pi}{4}$

80. Volumes.

(a) $\frac{\pi}{3}(4\sqrt{2} - \frac{7}{2})$ (b) 32π

81. Centre of Mass.

(a) $(\frac{11}{18}, \frac{65}{126})$ (b) $(0, 0, \frac{7}{3})$

82. Averages.

(a) $\frac{7}{6}$ cm (b) (i) 32 (ii) on $x^2 + y^2 + z^2 = 1$

83. Moment of Inertia.

$$\frac{8\pi\rho R^5}{15}$$

84. Another Application.

$\$7\left(\frac{\pi^4}{3} + 4\pi^2\right) \approx \$503.64$

MAST20009 Vector Calculus

Problem Sheet 5

Integrals over Surfaces

85. Surface Area.

(a) $\frac{\pi}{3}(6\sqrt{6}-8)$ (b) $4\pi^2 R$

86. Surface Integrals of Scalar functions.

(a) πa^3 (b) $\frac{1}{\sqrt{6}}$ (c) $\frac{40}{3}$

87. Mass of Sphere.

$\frac{4}{3}\pi R^4$

88. Flux Across a Surface.

(a) $\frac{12\pi}{5}$ (b) π (c) 4π

89. Surface Integrals of Vector fields.

(a) 24 (b) -16π

MAST20009 Vector Calculus

Problem Sheet 6

Integral Theorems

90. Green's Theorem.

(a) 0 (b) $-\pi R^2$

91. Apply Green's Theorem.

0, 8

92. Area via Line Integrals.

(a) $3\pi a^2$ (b) πab

93. Harmonic Functions.

94. Divergence Theorem in Plane.

(a) 2π (b) 5

95. Apply Divergence Theorem in Plane.

(a) 24 (b) 0

96. Integral Theorem Proof.

97. Stokes' Theorem.

(a) 0 (b) 4π

98. Apply Stokes' Theorem.

(a) -16π (b) 2π (c) $\frac{25\pi}{8}$

99. Integral Proofs.

100. Conservative Fields.

(a) $\phi = x^2yz - \cos x + c,\ 38 - \cos 3 + \cos 1$ (b) $\phi = xz^3 + x^2y + c,\ 202$

101. Further Line Integrals.

(a) $e \sin 1 + \frac{1}{3}e^3 - \frac{1}{3}$, (b) $-\frac{3}{2}$

102. Divergence Theorem.

3

103. Apply Divergence Theorem.

(a) $\frac{12\pi}{5}$, 4π (b) $\frac{3\pi}{4}$

104. More Integral Proofs.

105. Mixed Integral Theorems.

(a) 20π (b) $\frac{\pi}{3}$ (c) $\frac{-3\pi}{2}$ (d) 27 (e) 0 (f) 0

106. Think Carefully.

(a) $\frac{-8\pi}{5}$ (b) 12π (c) -2π

107. Maxwells Equations.

MAST20009 Vector Calculus

Problem Sheet 7

General Curvilinear Coordinates

108. Elliptical Cylindrical Coordinates.

(a) $\mathbf{e}_u = \dfrac{(\sinh u \cos v, \cosh u \sin v, 0)}{\sqrt{\sinh^2 u \cos^2 v + \cosh^2 u \sin^2 v}}, \quad \mathbf{e}_v = \dfrac{(-\cosh u \sin v, \sinh u \cos v, 0)}{\sqrt{\sinh^2 u \cos^2 v + \cosh^2 u \sin^2 v}},$
$\mathbf{e}_z = (0, 0, 1)$

109. Cartesian to Cylindrical.

$(z \cos\phi - \rho \sin 2\phi)\hat{\boldsymbol{\rho}} - (z \sin\phi + 2\rho \cos^2\phi)\hat{\boldsymbol{\phi}} + \rho \sin\phi \hat{\boldsymbol{z}}$

110. Cylindrical Coordinates.

(a) $\frac{1}{\rho}\hat{\boldsymbol{\phi}}$ (b) $n\rho^{n-1}\hat{\boldsymbol{\rho}}$ (c) $3\cos\phi$ (d) 4 (e) $\frac{1}{\rho}\hat{\boldsymbol{z}}$ (f) 0

111. Spherical Coordinates.

(a) $\dfrac{1}{r\sin\theta}\hat{\boldsymbol{\phi}}$ (b) $\dfrac{1}{r}\hat{\boldsymbol{\theta}}$ (c) $\dfrac{2\cot\phi}{r}$ (d) $\dfrac{-2\cot\theta}{r}\hat{\boldsymbol{r}} + \left(\dfrac{2}{r} - \dfrac{\operatorname{cosec}^2\phi}{r\sin\theta}\right)\hat{\boldsymbol{\theta}}$
(e) $\dfrac{2}{r} + \dfrac{\cot\theta}{r^2}$ (f) $\dfrac{2\cos\theta}{r}$